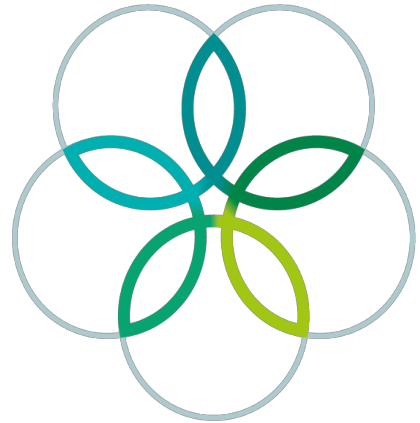
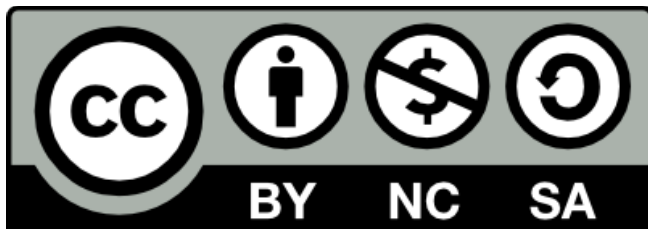


INTERNATIONAL
BIOLOGY
OLYMPIAD e. V.

IBO



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Eckhard Klieme
Eckhard R. Lucius
Erwin Zabel (Eds.)

Report of the 9th International Biology Olympiad

ipn-materialien



Horst Binding, Christian Gliesche, Ludger Kappen, Eckhard Klieme,
Ralf Kittler, Sievert Lorenzen, Eckhard R. Lucius, Erhard Lipkow,
Karl Meissner, Hansjörg Rudolph, Jürgen Soll, Helmut Uhlarz, Erwin Zabel

4 The Tasks of the 9th International Biology Olympiad

5.1 Theoretical Tests

5.2.1 Theoretical Test - Part A

01-10	1 B	2 C(2)	3 B	4 A(2)	5 B	6 A(2)	7 D	8 D	9 B	10 A
11-20	11 C	12 D	13 C(2)	14 A	15 A	16 D	17 A	18 B(2)	19 D(2)	20 C
21-30	21 D	22 B(2)	23 D	24 D	25 C	26 B	27 D	28 D	29 D	30 C
31-40	31 A	32 A	33 B	34 B(2)	35 D	36 A	37 C	38 A	39 C(2)	40 B
41-50	41 B(2)	42 D	43 C(2)	44 C	45 D	46 D	47 C(2)	48 C	49 E	50 B
51-60	51 B	52 D	53 C	54 B(2)	55 C(2)	56 D(2)	57 D	58 D	59 E	60 D
61-70	61 E	62 D	63 B	64 D(2)	65 B(2)	66 C	67 B	68 C	69 B	70 B
71-80	71 D(2)	72 C	73 C	74 A(2)	75 A	76 C	77 B	78 C	79 E(2)	80 C

Tab. 11: Answer codes of the following tasks, part A of the theoretical test

The following part A tasks were skipped: 2, 5, 11, 29, 33, 39, 43, 49, 48.

CELL BIOLOGY

A 1: How do polypeptides find their way from the place of synthesis (1 pt) at cytoplasmic ribosomes to the place of their destination in the mitochondria?

- A: by specific transport along the cytoskeleton
- B: by specific amino-terminal targeting signals
- C: by specific carboxy-terminal targeting signals
- D: it is not necessary because the synthesis takes place on the surface of the organelles
- E: mitochondria synthesize all proteins inside the organelle

A 2: What is mRNA editing? (2 pts)

- A: removal of a coding base triplet from the mRNA
- B: alteration of the coding sequence during transcription
- C: change of a single base in the mRNA
- D: addition of a poly-adenylat-tail to the mRNA
- E: reverse transcription from mRNA to DNA

A 3: How does the number of mitochondria and plastids (1 pt) increase in a cell?

- A: only by de novo synthesis
- B: only by division
- C: by de novo synthesis and division
- D: by inheritance
- E: by fusion of membrane vesicles

A 4: How is the distribution of light to the photosystem I (PS I) and (2 pts) the photosystem II (PS II) adjusted to different conditions of light?

- A: by changes in the size of antenna, in the orientation of the chloroplasts or in the thickness of leaves
- B: by reversible removal of polypeptides of the reactive centre, and un packing of grana thylakoids
- C: by separation of chlorophyll from PS I and PS II
- D: by inhibiting the electron flow from PS I to PS II
- E: by uncoupling the ATP synthesis

A 5: Which process initiates the proteolytic degradation of proteins?
(1 pt)

- A: glycosylation C: prenylation
B: ubiquitination D: phosphorylation E: ADP ribosylation

A 6: How do you experimentally differentiate between integral and peripheral membrane proteins?
(2 pts)

- A: Only peripheral proteins (but not integral) can be removed from the membrane by 1 M NaCl solution.
B: Only integral membrane proteins can be removed from the membrane by 0.1 M NaOH.
C: Only integral membrane proteins can be removed from the membrane by 1 M NaCl.
D: Only peripheral (but not integral) membrane proteins can be solubilized in detergent.
E: Only peripheral (but not integral) membrane proteins can be affected by proteases.

A 7: How are the fibre types forming the cytoskeleton called?
(1 pt)

- A: tubuline, lignin, kinesin
B: microtubules, myosin, microfilaments
C: keratin, myosin, kinesin
D: microfilaments, intermediate filaments, microtubules
E: actin, myosin, intermediate filaments

A 8: Animal and plant cells possess channels directly connecting (1 pt) the cytoplasm of one cell to the cytoplasm of another cell. How are these structures called?

- A: plasmodesmata, desmosomes
B: plasmodesmata, Ca²⁺-ATPase
C: porin, gap junction
D: gap junction, plasmodesmata

A 9: What is the endosymbiosis hypothesis related to?

(1 pt) I. to bacteria living in the epithelium of gizzard

II. to plastids III. to mitochondrial IV. to ribosomes

A: I

B: only II and III

D: only III and IV

C: only III

E: II, III and IV

A 10: The semipermeability of which of the water-permeable (1 pt) membranes would be sufficient for the process of plasmolysis?

I. plasmalemma

II. tonoplast

III. each biomembrane enclosing a compartment

IV. the middle lamella

A: only I C: I and II

B: only II D: only III E: III and IV

A 11: Which process are microtubules not involved in?

(1 pt)

A: cytomorphogenesis C: mitotic prophase

B: meiotic anaphase I D: transport of Golgi vesicles

E: cytoplasmic circulation

A 12: A B-lymphocyte (a cell type of the immune system) produces (1 pt) and secretes antibodies. Which structures of its protoplast should therefore be very well developed?

A: only the smooth endoplasmic reticulum

B: only the smooth endoplasmic reticulum and the Golgi apparatus (dictyosomes)

C: only the rough endoplasmic reticulum and the lysosomes

D: only the rough endoplasmic reticulum and the Golgi apparatus

E: the rough endoplasmic reticulum, the Golgi apparatus and the lysosomes

A 13: Which of the following reactions can occur in the mitochondria?
(2 pts)

- | | |
|------------------------------|---|
| I. NADP reduction | V. Calvin |
| II. synthesis of fatty acids | VI. citric acid cycle |
| III. endoxidation | VII.oxidative catabolism of fatty acids |
| IV. gene expression | VIII. nitrit reduction |

- A: I, III, VI, VII
 B: II, III, IV, VII
C: III, IV, VI, VII
 D: IV, VI, VII, VIII
 E: I, III, V, VIII

A 14: Which of the following are found in the cell wall of flowering plants?
(1 pt)

- | | | |
|-------------------|------------------------|------------|
| I. apoplast | IV intermicellar space | VII lignin |
| II. mircofilament | V keratin | VIII cutin |
| III. microfibre | VI chitin | IX suberin |

- A: I,III,VII,IX,
 B: II,V,VII,VIII,
 C: III,IV,V,VIII
 D: II,VI,IX
 E: I,III,V,VI

A 15: What is a prion?

(1 pt)

- A: a protein
 B: infectious RNA without genes for a protein envelope
 C: the DNA sequence which serves as a template for primer RNA
 D: an early eukaryote free of mitochondria
 E: a multienzyme complex for the biosynthesis of fatty acids

A 16: Which statements concerning the growth factors cytokins and (1 pt)cytokinins are correct?

- I. The plant cytokinins are peptides.
- II. The plant cytokins are peptides.
- III. The plant cytokinins are purine derivatives.
- IV. The plant cytokins are purine derivatives.
- V. The animal cytokinins are peptides.
- VI. The animal cytokins are peptides.
- VII. The animal cytokinins are purine derivatives.
- VIII. The animal cytokins are purine derivatives.

- A: I and VII
- B: II and VIII
- C: II and V
- D: III and VI
- E: IV and VI

A 17: Which classes of lipids have non-polar side chains and polar (1 pt)head groups?

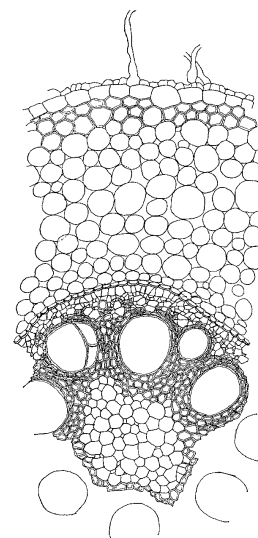
- A: phospholipids
- B: triglycerides
- C: cholesterol
- D: waxes
- E: glycerol

ANATOMY AND PHYSIOLOGY OF PLANTS

A 18: The figure below shows the cross-section of a plant or organ. (2 pts) **Distinguish** which of the given morphological, anatomical, systematical and ecological terms are in accordance with the cross-section. **Select a combination of possibilities.**

- I. stem axis
- II. root
- III. leaf stalk
- IV. dicotyledon
- V. monocotyledon
- VI. swamp or water plant
- VII. terrestrial plant

- A: I, IV, VI C: II, V, VI
B: II, V, VII D: II, IV, VII
 E: III, IV, VII



A 19: In the following three plant species(I to III) are listed. Among 1 to 7 (2 pts) **anatomical and physiological characteristics** are mentioned as well.

I. <i>Opuntia ficus-indica</i> (Cactaceae)	1. water storage tissue of cells with large vacuoles and chloroplasts
II. <i>Zea mays</i> (corn, Poaceae)	2. chloroplasts of bundle sheath cells with reduced grana.
III <i>Triticum aestivum</i> (wheat, Poaceae)	3. C ₃ -plant
	4. C ₄ -plant
	5. CAM-plant
	6. subepidermal sclerenchyma in the
	7. stomata predominantly open at night

Decide which of the following assignments are correct.

- A: I: 1, 3, 6 II: 1, 5, 7 III: 2, 4, 6
 B: I: 2, 4, 6 II: 3, 6 III: 1, 3, 6
 C: I: 2, 4, 6 II: 2, 3, 6 III: 3, 6

D: I: 1, 5, 7 II: 2, 4, 6 III: 3, 6

E: I: 1, 5, 7 II: 3, 6 III: 2, 4, 6

A 20: A C₃-plant and a C₄-plant are kept together under an airtightly (1 pt)sealed glass bell under light. How does the CO₂-concentration change under this glass bell?

A: The CO₂-concentration does not change.

B: The CO₂-concentration decreases to the CO₂-compensation point of the C₃-plant.

C: The CO₂-concentration decreases to the CO₂-compensation point of the C₄-plant

D: The CO₂-concentration increases.

E: The CO₂-concentration drops below the value of the CO₂-compensation point of the C₄-plant.

A 21: The protoplast has detached from the cell wall after plasmolysis. (1 pt) What can mainly be found between the cell wall and the protoplast?

A: air

B: vacuum

C: water

D: hypertonic solution

E: cell sap

A 22: What happens if you put a turgescient cell into its 'own cell sap' (2 pts) (i.e. a solution with the same water potential as the osmotic potential of the cell)?

A: There is no change.

B: The cell loses water until its water potential is equal to the water potential of the surrounding solution.

C: Water is released until incipient plasmolysis occurs.

D: Plasmolysis occurs.

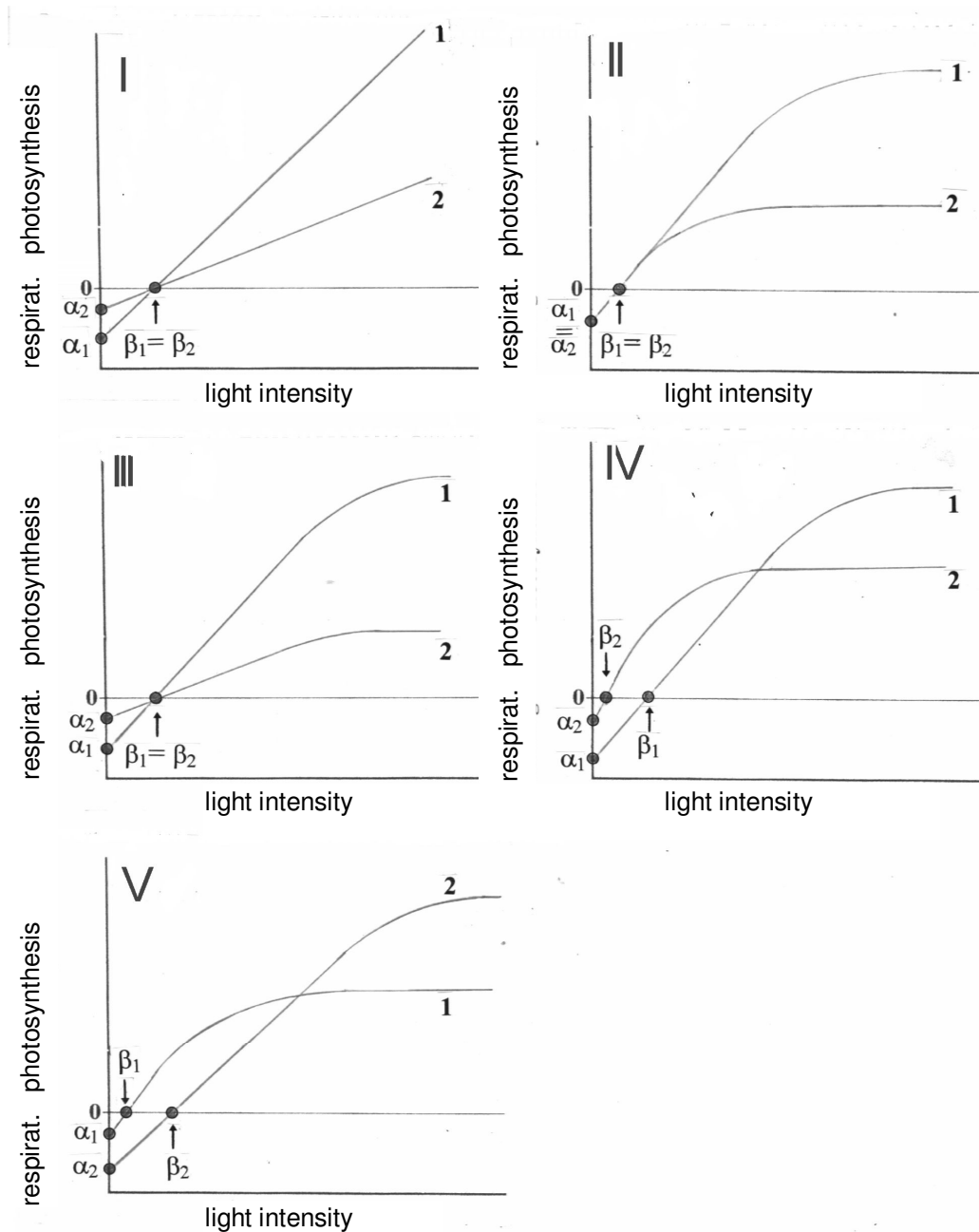
E: The cell bursts.

A 23: By examining the dependence of photosynthesis on photon supply you get so-called light curves (curves I to V).

(1 pt)

Legend: 1: typical sun plant; 2: typical shade-loving plant;

α : respiration in darkness; β : light compensation point

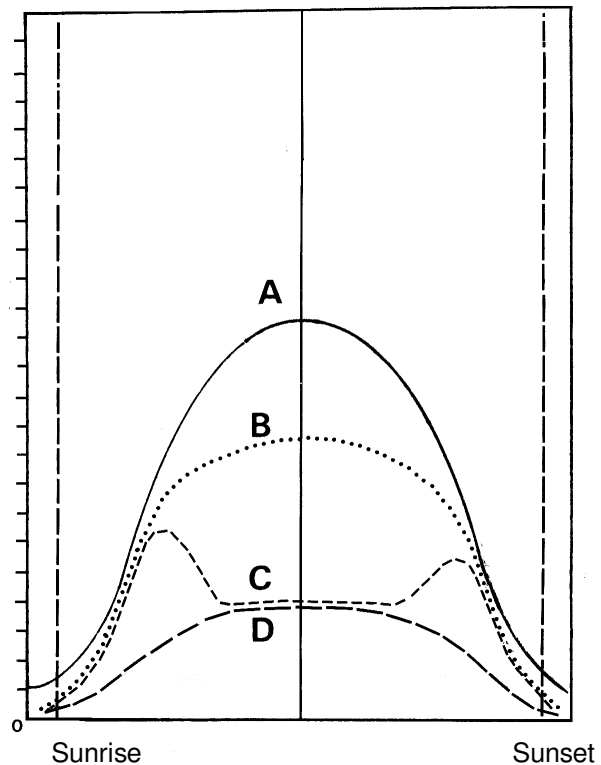


Which of the preceding curves including the inscription is correct?

A: I B: II C: III D: IV E: V

A 24: The figure shows - besides evaporation the daily curves for the process of transpiration of a plant underlying deterioration of water supply. Which of the curves expresses the daily course of the transpiration in the cuticle?

- A: curve A
- B: curve B
- C: curve C
- D: curve D



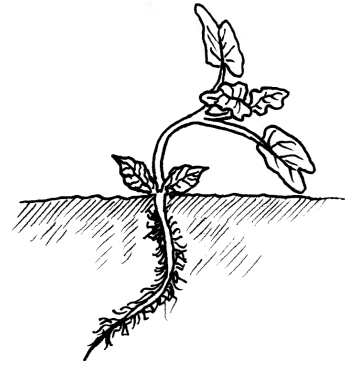
A 25: If a respiratory quotient > 1 is measured in an experiment, that (1 pt) means:

- A: In the measured respiration process carbohydrates are used as substrates.
- B: In the measured respiration process fats are used as substrates.
- C: In the measured respiration process organic acids are used as substrates.
- D: In the measured respiration process proteins are used as substrates.

A 26: The following figure shows a germ which was grown in hydro-

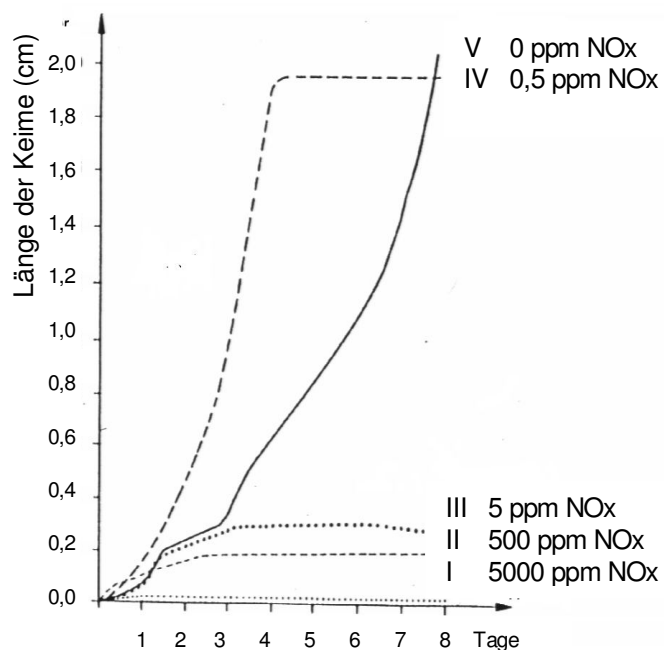
(1 pt) **ponics in a glass vessel and which was exposed to light only from the right-hand side. The growth curvatures you can observe in the stem axis are called:**

- A: photonasty
- B: phototropism
- C: phototaxis
- D: photomorphosis
- E: photoperiodism



A 27: Germs were grown in closed chambers (I-IV) with different (1 pt) **nitrogen oxide NO_x -concentrations in the air. The following figure demonstrates the average length of the germs depending on time. The nitrogen oxide concentration (kept constant) is given in ppm together with the corresponding chamber number.**

- A: The lower the NO_x -concentration the larger the germs can grow on average within 8 days.
- B: In an atmosphere free of NO_x germs grow faster from day 5 to 8 than in the first four days.
- C: For achieving an average germ length of 1.5 cm half of the time that would be needed



without NO_x is sufficient in a concentration of 0.5 ppm NO_x.

- D: At a concentration of 5000 ppm there is nearly no germ growth, but 1/10 of this concentration promotes growth during the fourth day.
- E: After 4 days in chambers with air containing at least 0.5 ppm NO_x, no germ growth can be found any more.

A 28: Which function does the glyoxylate cycle have during (1 pt) germination?

- A: reduction of fatty acids
- B: synthesis of ATP
- C: decarboxylation of glyoxylate from photorespiration
- D: linking the β -oxidation to gluconeogenesis
- E: synthesis of glycine by transamination

A 29: Which three sensory systems regulate light-depending development and differentiation of angiosperms? (1 pt)

- I. phytochrome
- II. cryptochrome
- III. protochlorohyllid
- IV. chlorophy
- V. anthocyan

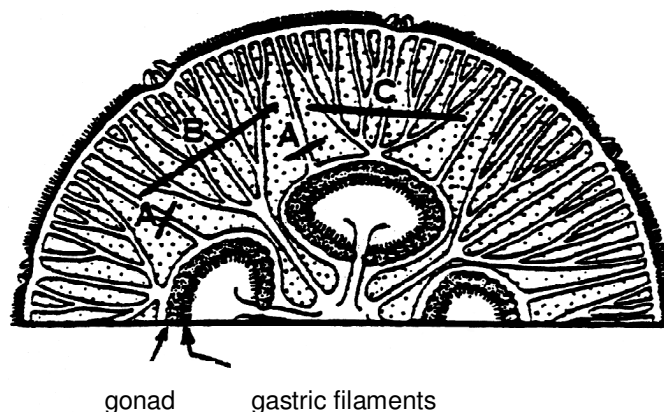
- A: I, III and V
- B: I, II and IV
- C: II, III and V
- D: I, II and III
- E: II, IV and V

ANATOMY AND PHYSIOLOGY OF ANIMALS AND MEN

A 30: The eyes of squids are highly efficient, those of snails are poorly efficient. Consider whether efficiency of organs may be coupled with the intensity of their metabolism. If squids were provided with poorly efficient eyes and snails with highly efficient eyes whereas the further body organisation remained unchanged in both cases, this would be:

- A: a disadvantage to squids and an advantage to snails
- B: an advantage to squids and a disadvantage to snails
- C: a disadvantage to both squids and snails
- D: an advantage to both squids and snails

A 31: A medusa metabolizes most energy at the rim of its umbrella. (1 pt) Energy is extracted from small particles which can be phagocytized on their whole way through the tubes of the gastrovascular system. How do the small particles have to be transported through the gastrovascular system of the moon jelly (see figure below) in order to provide the rim of its umbrella with a maximum amount of food?



- A: through tubes marked by A towards the rim of the umbrella
- B: through tubes marked by B towards the rim of the umbrella
- C: through tubes marked by C towards the rim of the umbrella
- D: no preference of particular tubes

A 32: In a smooth muscle, the length difference between its relaxed and (1 pt) its maximally contracted state is

- A: larger than in a striated muscle
- B: smaller than in a striated muscle
- C: as large as in a striated muscle
- D: null, because smooth muscles cannot contract themselves but, rather, prevent tissues from becoming stretched.

A 33: Let striated skeletal muscles be replaced by smooth ones, and, (1 pt) reversely smooth muscles of the intestinal wall by striated ones. The effect would be:

- A: Both locomotion and intestinal peristaltic would be improved.
- B: Both locomotion and intestinal peristaltic would be impaired.
- C: Both locomotion and intestinal peristaltic would be as before.
- D: Locomotion would be impaired, but intestinal peristaltic would be improved.
- E: Locomotion would be improved, but intestinal peristaltic would be impaired.

A 34: Many species of *Rotatoria* (rotifers; up to 3 mm length) in (2 pts) freshwater have a reproductive cycle called heterogony: Reproduction occurs temporarily only in females and temporarily in males and females. If this is an advantage to these species of *Rotatoria*, which of the following statements is correct?

- A: Several generations of females and males are followed by one with only females.
- B: Several generations of only females are followed by one with females and males.
- C: Every generation of only females is followed by one with females and males, and vice versa.
- D: Several generations of only females are followed by several with females and males, and vice versa.

A 35: Stamina training leads to many effects in man. Which of the (1 pt) following effects does not apply to stamina training?

- A: enlargement of the heart
- B: increase of the number of capillaries in muscles
- C: increase of the number of mitochondria within muscle cells
- D: increased pulse rate during rest
- E: increase of the lung volume

A 36: Manual work increases the CO₂ concentration in the blood. What (1 pt) is the result of this?

- I. The affinity of hemoglobin to oxygen decreases.
- II. Elimination of CO₂ through the lungs decreases.
- III. Loss of mineral salts through the air breathed out increases.
- IV: The brain suffers from oxygen deficiency.
- V: The body warms up.

A: only I B: only IV C: I, V D: II, III E: II, IV

A 37: What short-term effects may occur in the human cardiovascular (1 pt) system at a sudden transition from lying to standing?

- I. The pulse rate is increased.
- II. The total peripheral resistance is increased.
- III. More blood flows through the kidneys.
- IV. More blood flows through the upper and lower limbs.
- V. The blood pressure is decreased.

- A: I, III, IV
- B: III, IV, V
- C: I, II, V
- D: only II, III
- E: only IV, V

A 38: If a fish species from fresh water developed into a marine fish species, which problems of osmoregulation would have to be solved?

- (1 pt)
- I. Excess of salts would have to be excreted.
 - II. Water would have to be reabsorbed from the initial ultrafiltrate of kidneys.
 - III. The salt concentration of the blood would have to be adjusted to that of the sea.
 - IV. Drinking of sea water would have to be avoided.
 - V. Permeability of the skin would have to be reduced.

A: only I, II

C: all answers

B: only II, IV

D: only V

E: only II, IV

A 39: In an everse eye (the opposite of inverse), the light-sensitive region of retinal cells is directed towards the incoming light. The figure below taken from a text book illustrates how an everse median eye of a spider could have developed by invagination. Why should this suggested eye structure be considered incorrect?

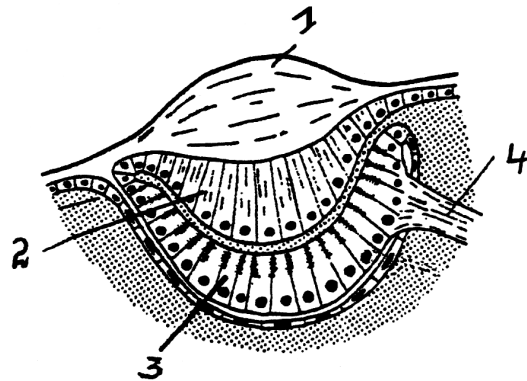
Legend:

1: lens

2: vitreous body

3: retinal cell, the sensory area is opposite to nucleus

4: optic nerve



A: In the development of eyes, invagination never plays a role.

B: The figure really shows an inverse eye.

C: Generally, sensory cells perceive the stimulus distally and conduct the impulses proximally.

D: Generally, sensory cells perceive the stimulus proximally and conduct the impulses distally.

E: The cells of the vitreous body and the cells of the retina are mixed up.

A 40: Bones are dynamic rather than static structures. How may this be (1 pt) realized?

- A: Those who sit crookedly will soon get a crooked back.
- B: In case of being bedridden, the internal structure of bones adapts to the altered stress.
- C: Bones do not break as readily as static structures of equal strength.
- D: A tooth crown freshly implanted, which initially does not fit well, can do so after a couple of weeks without intervention of the dentist.

A 41: If a man of 70 kg weight ingests 40 g alcohol, the alcohol level in (2 pts) his blood will raise one part per thousand. About 1 g alcohol is eliminated per hour and 10 kg body weight. The man of 70 kg body weight has been involved in a traffic accident and has run away. A blood sample has been taken from him after 2 ½ hours. It contained 0.5 parts per thousand of alcohol. Assuming he did not ingest any alcohol after the accident - how much alcohol did his blood contain at the time of the accident?

- A: 1.10 parts per thousand
- B: 0.95 parts per thousand
- C: 0.80 parts per thousand
- D: 0.65 parts per thousand
- E: 0.55 parts per thousand

ETHOLOGY

A 42: The mobbing reaction of house swallows (*Hirundo rustica*). (1 pt)



House swallows mostly breed in colonies. They live on flying insects. Predators spotted during flights for prey or near the nest are noisily attacked in nose-dives. This mobbing reaction communicates itself to the other swallows and supports the defence against enemies. The table shows the ethological status of the swallows and their share in the mobbing reaction and in the population, respectively.

birds' status	share in the population in per cent	share in the mobbing reaction in per cent
adult animals, not mated	6	2
adult animals, before breeding	9	11
adult animals, breeding	14	10
adult animals with young animals	51	77
young animals	20	0
total	100	100

Which of the following hypothesis have to be assumed to be correct according to the results?

- I: The mobbing reaction is an intra-specific signal.
- II: The mobbing reaction is an element of self-defence.
- III: The mobbing reaction is an element of choice of mates.
- IV: The mobbing reaction is a kind of maternal care.
- V: The mobbing reaction is a form of commensalism.
- VI: The mobbing reaction is an altruistic mode of behaviour.

A: IV, V, VI B: III, IV, V C: II, IV, VI D: I, IV, VI

A 43: The rank of macaques (*Macaca sylvanus*).

(2 pts)

Macaques are related to rhesus macaques; they live in Morocco and Algeria. Adult males, females and their young animals form groups in which ranks are set up by superorder or suborder.

The table below shows the frequencies of aggression within a group of nine females (A to I):

being attacked

	A	B	C	D	E	F	G	H	I	total
A	-	16	6	6	16	-	4		5	53
B	-	-	4	14	10	-	2	1	2	33
C	-	-	-	7	8	1	2	-	5	23
D	-	-	-	-	18	2	-	1	8	29
E	-	-	-	-	-	7	5	10	11	33
F	-	-	-	-	-	-	2	14	8	24
G	-	-	-	-	-	-	-	10	-	10
H	-	-	-	-	-	-	-	-	6	6
I	-	-	-	-	-	-	13	-	-	13
	0	16	10	27	52	10	28	36	45	224

attacking

Read the matrix and choose which one of the females is the lowest in the rank order.

How do you call the individual losing conflicts within the group?

- | | | |
|-------------------|--------------------|-----------------|
| I. female A | II. female I | III. female E |
| IV. female F | V. female H | VI. female G |
| VII. omega animal | VIII. alpha animal | IX. subdominant |

Which of the following combinations is the correct one?

- | | |
|------------------------|---------------|
| A: I, III, IV | D: IV, VI, IX |
| B: II, V, VII | E: V, VII, IX |
| <u>C: III, VII, IX</u> | |

GENETICS/EVOLUTION

A 44: Which of the following statements on introns is correct?
(1 pt)

- A: They are non-transcribed sequences (spacers) between two genes.
- B: They are transcribed spacers between two genes.
- C: They are located between the coding regions of a gene.
- D: They are located between the coding regions of a mature mRNA.
- E: They are non-coding regions of a polycistronic mRNA.

A 45: Which statements on sex chromosomes (gonosomes) are correct?
(1 pt)

- I. They are always heterochromatic.
- II. They may occur in a euchromatic or heterochromatic form.
- III. They occur in human beings and in all animals.
- IV. They occur in yeast (*Saccharomyces cerevisiae*).
- V. They occur in some mosses.
- VI. They occur in some flowering plants.
- VII. Their homozygosis always leads to female sex.

- A: I, II, III C: I, IV, VI
B: I, III, V D: II, V, VI E: II, V, VII

A 46: The protoplast fusion in dicotyledons is well suited for the following objectives A-D. What can also be achieved by cross hybridization?
(1 pt)

- A: combination of the genetic material of different plant families
- B: studies on the fate of plastids during the development from cells with mixed plastide populations
- C: studies on the fate of mitochondria during the development from cells with mixed mitochondria populations
- D: transfer of resistance genes from wild species into cultivated species

A 47: Did Gregor Mendel know some of the phenomena (I- VII)? Under (2 pts) which capital letter do you only find such phenomena?

- I. decrease of the degree of homozygous-recessives in the generation sequence
- II. extrakaryotic inheritance
- III. break of coupling
- IV. polyploidy
- V. hybrid character of the endosperm (albumens)
- VI. cleistogamy in pea
- VII. homogeneity of the F1 of homozygous parents differing in two pairs of alleles

A: I, II C: V, VI E: IV, VII
B: I, III D: III, VI

A 48: Which of the following statements characterizes a ‘substitution (1 pt)hybrid’?

- A: The hybrid of two auxotrophic mutants is prototrophic
- B: The deficiency caused by a recessive allele is repaired by another gene
- C: A pair of chromosomes is replaced by a pair of chromosomes of another species.
- D: The hybrid competes one of the parents out of the population
- E: The plastids of one species are displaced by those of another species (after back-crossing).

A 49: Which process does not lead to a change in the chromosome (1 pt)number?

- A: poisoning of the nuclear spindle
- B: non disjunction
- C: endomitosis
- D: centromere fracture
- E: duplication

A 50: The bacteria strains A and B and a certain bacteriophage which is (1 pt)infectious for both strains grow in one culture. After one day some bacteria which are equivalent to type A have originated, but show characteristics based on five alleles in B.

Which processes might have caused this phenomenon?

- | | |
|-------------------|------------------|
| I. transformation | IV. sexduction |
| II. transposition | V. mutation |
| III. transduction | VI. transfection |

A: I, II, III C: I, V, VI E: , III, IV

B: I, III, IV D: II, IV, VI

A 51: The table sums up characteristics which are original (plesiomorphic) or deduced (apomorphic) within the group of angiosperms. (1 pt)

	characteristics
1	vascular bundle collateral closed, herb
2	vascular bundle collateral open, woody plant
3	blossom polysymmetric, actinomorphic
4	blossom monosymmetric, zygomorphic
5	monospermous indehiscent fruit (e.g. nut)
6	polyspermous dehiscent fruit (e.g. capsula)
7	tracheal segments, long
8	tracheal segments, short
9	blossoms without nectar
10	blossoms with nectar
11	ovary superior
12	ovary inferior

Which of the following combinations shows deduced (apomorphic) characters only?

A: 1, 3, 5, 7, 10, 11 C: 2, 4, 6, 8, 9, 12

B: 1, 4, 5, 8, 10, 12 D: 2, 3, 5, 8, 9, 12 E: 1, 4, 6, 7, 10, 12

A 52: I - V lists names of plants and 1 - 5 lists some characteristics (1 pt)referring to the reproduction of cormophytes.

- I. *Lycopodium (Lycopodiophyta)*
- II. *Gingko (Pinophyta)*
- III. *Pinus (Pinophyta)*
- IV. *Cycas (Pinophyta vel Cycadophyta)*
- V. *Magnolia (Magnoliophyta)*
 - 1. aggregations of sporophylls (strobili) or blossoms unisexual
 - 2. aggregations of sporophylls (strobili) or blossoms bisexual
 - 3. aggregations of sporophylls (strobili) or blossoms sexually not differentiated
 - 4. fertilization by spermatozoids
 - 5. fertilization by immobile sperms

Which assignments are correct?

- | | | | | | |
|-----------|---------|----------|-----------|----------|---------|
| A: | I: 1, 4 | II: 3, 4 | III: 3, 4 | IV: 1, 5 | V: 3, 4 |
| B: | I: 1, 5 | II: 2, 5 | III: 1, 4 | IV: 2, 5 | V: 2, 5 |
| C: | I: 3, 4 | II: 1, 4 | III: 1, 5 | IV: 2, 5 | V: 2, 5 |
| <u>D:</u> | I: 3, 4 | II: 1, 4 | III: 1, 5 | IV: 1,4 | V: 2, 5 |

A 53: Which advantage do angiosperms (with double fertilisation and (1 pt)formation of a secondary endosperm) have in comparison to gymnosperms (single fertilisation and formation of a primary endosperm)?

- 1. The neotenic development of a female prothallium (macrogametophyte) requires less energy.
- 2. The triploid endosperm of angiosperms can store more nutrients.
- 3. Regarding the investment of energy and reserve material, the formation of a secondary endosperm after double fertilization is more economic.
- 4. Double fertilization is an additional prevention against hybridization.
- 5. Double fertilization accelerates the embryonic development.

- | | | | |
|----|-----------|-----------|-----------|
| A: | 1, 2, 3 | <u>C:</u> | only 1, 3 |
| B: | only 2, 4 | D: | 2, 4, 5 |
| | | E: | 1, 3, 4 |

A 54: Lamarck and Darwin each proposed a theory on the origin of (2 pts) species diversity. According to Lamarck's theory of species transformation, 'infusories' may originate and have to originate by abiogenesis up to the present and give rise to the development of higher organisms. A single origin of organisms is sufficient for Darwin's theory of natural selection. Which of the following statements is correct?

- A: Lamarck has to be considered as the author of the descent theory which, however, was first founded by Darwin.
- B: Different contemporary species may be related to each other according to Darwin's theory, but not according to Lamarck's theory.
- C: Long-term evolving of species depends on improvement in performance of organisms according to Darwin's theory and on increasing success of organisms according to Lamarck's theory.
- D: Lamarck considered inheritance of acquired characteristics as possible, whereas Darwin did not.

A 55: The well known evolutionary geneticist Maynard Smith stated in (2 pts) 1989: 'We need some measure of the survival and reproduction of the different genotypes. This measure is fitness.' Which of the following statements is correct?

- A: While organisms are mortal, genes are potentially immortal. Therefore, the statement cited is correct.
- B: Indeed, fitness is a condition to achieve various kinds of success. Fitness can only be measured by measuring the latter. Therefore, the statement cited is correct.
- C: According to the statement cited, the success of survival and reproduction is a measure of fitness. However, fitness is a condition to achieve success. Therefore, the statement cited is wrong.
- D: Fitness of genotypes can only be measured by the survival and reproductive success of the corresponding organisms. Therefore, the statement cited is correct.
- E: The suggested measure of fitness works well. Therefore, it does not matter whether the statement cited is right or wrong.

ECOLOGY

A 56: Why do some columnar cacti in their natural habitat turn their (1 pt)longitudinal axis to the point where the sun is at its zenith in summer-time? Evaluate the explanations I - III.

- I. to attain as much light as possible
- II. to heat the apically arranged reproductive organs in the morning and in the evening
- III. to minimize the increase in temperature of the columnar body

Which of this / these explanation(s) is (are) correct?

- A: None of the explanations is correct.
- B: Only explanation I is correct.
- C: Only explanations I and II are correct.
- D: Only explanation III is correct.
- E: All three explanations are correct.

A 57: Which plant life form of the soil flora in a deciduous forest is (1 pt)developing and fruiting most quickly in spring?

- A: therophyts, because they have seeds that already germinate at low temperatures
- B: therophyts, because their subterranean storage organs allow a quick start of growth
- C: geophyts, because their seeds quickly germinate on the soil in spring
- D: geophyts, because they can quickly form a shoot from their subterranean storage organs
- E: None of the explanation above is correct.

A 58: In which way does a cormophytic plant protect itself against (1 pt)freezing ($< 10^{\circ}\text{C}$)?

- A: Plants develop a dense branching to prevent the loss of heat.
- B: Plants protect themselves against freezing by a thick cuticle and thick hair on the leaves.
- C: Plants concentrate salts in their cells (osmotic effect).
- D: Plants have to tolerate cellular dehydration as a consequence of ice-formation.
- E: Freezing is avoided by metabolic heat production.

A 59: Why do you often find cyanobacteria and red algae in places with (1 pt) low irradiance?

- I. All of them are small and are outcompeted by fast-growing species.
- II. They are strongly photosensitive.
- III. They can efficiently utilise light with the help of additional antenna pigments

- A: All three explanations are correct.
B: Only I is correct.
C: Only II is correct.
D: Only III is correct.
E: Only II and III are correct.

A 60: Why is a high-moor bog an extreme habitat? (1 pt)

- I. Because it is oligotrophic and permanently cold.
- II. Because it allows extreme situations on the surface concerning climate.
- III. Because it is oligotrophic and has a low pH value.

- A: I, II and III C: I and II E: Only III
B: Only I D: II and III

A 61: Which one of the desert plants I and II has to tolerate more heat in (1 pt) a habitat with the same solar radiation and why?

- I. a desert plant with large leaves and water-storing bulbs
- II. a desert plant with small but succulent leaves

- A: I heats up more intensively than II, because the leaves possess a large surface.
B: I and II heat up at the same range and have to be equally heat tolerant.
C: II heats up more intensively, because its heat capacity is low.
D: II heats up more intensively than I, because the leaves contain a lot of water.
E: II heats up more intensively than I, because the leaves of I transpire more intensively.

A 62. Rips are an advantage to columnar cacti and globular cacti:

(1 pt) I. because they are in agreement with the concept of xerophytes by increasing the surface.

II. because predators are irritated.

III. because they decrease the irradiation by partial shading

A: Only I

C: I and II

E: I, II and III

B: Only II

D: Only III

A 63: The ovary needs a lot of heat energy for maturation. In cold (1 pt) climates this may be a problem. By which means is the perianth capable of increasing the energy input in a particularly effective way?

A: by a bell shaped perianth preventing the nightly emission

B: by paraboloid flowers reflecting the solar radiation to the ovary

C: by dark-coloured flowers absorbing a particularly high amount of heat

D: by the symmetry of zygomorphic flowers causing an evenly warming up of the ovary

E: by an ovary which is bigger than the perianth

A 64: R-strategists (r-selectionists) are inhabitants of fast originating (2 pts) and vanishing habitats. They are selected for quick colonisation, for fast and complete utilisation of these habitats and for fast search for new favourable places. Which of the following characteristics apply to r-strategists?

I. They have a slow development.

II. They have a relatively large number of offspring.

III. They have a short life span.

IV. Their mortality is independent of the population density.

V. Their population size is relatively constant.

A: I, II and III

C: II, III and V

E: I, III and V

B: I, II and IV

D: II, III and IV

A 65: The ecosystems shown in the table differ in the amount of their (2 pts) **net primary production**.

selected ecosystems

number	ecosystem
1	tropical rain forest
2	savannah
3	subtropical sand desert
4	temperate deciduous forest
5	boreal deciduous forest
6	tundra

Name the correct order (increasing net primary production) of the ecosystems shown in the table.

- A: 3, 6, 2, 5, 4, 1
B: 3, 6, 5, 2, 4, 1
 C: 6, 3, 5, 2, 4, 1
 D: 6, 3, 2, 5, 1, 4

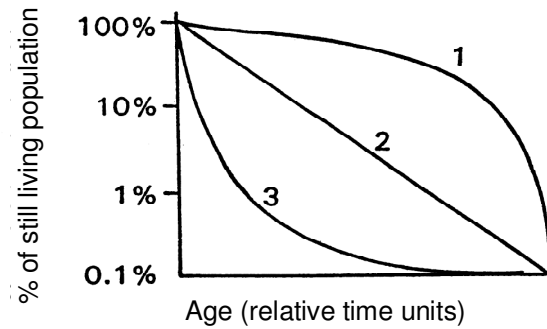
A 66: Which possible effects does the smoke from the vast forest fires (1 pt) **in tropical rain forest have on the plants in neighbouring areas?**

- I. decrease of the photosynthetic rate
- II. increase in the respiration rate
- III. decrease of the plant growth
- IV. increase in the transpiration rate

- A: I and II
 B: II and III
C: I and III
 D: III and IV
 E: II and IV

A 67: The following figure shows survivorship curves. Inspect the curves showing the percentage of individuals of populations of three different species which are still alive at any particular age.

- I. elephant
- II. frog
- III. rabbit



Inspect the survivorship curves. Which of the following combinations is correct?

- A: I (2), II (3)
- B: I (1), II (3)
- C: II (3), III (1)
- D: I (1), II (2)
- E: II (2), III (3)

SYSTEMATICS

A 68: Which characteristics are typical for wind-pollinated (anemophilous) plants? (1 pt)

- A: smooth stigma, viscous pollen and polyspermous dehiscent fruit
- B: smooth stigma, non-viscous pollen and polyspermous indehiscent fruit
- C: papillate stigma, non-viscous pollen and monospermous indehiscent fruit
- D: papillate stigma, viscous pollen and monospermous dehiscent fruit
- E: papillate stigma, viscous pollen and polyspermous indehiscent

A 69: Where do beans (*Fabaceae*, e.g. soybean *Glycine max*, syn. *Soja hispida*) store reserve material for germination? (1 pt)

- A: in the pericarp
- B: in the cotyledons of the embryo
- C: in the triploid nutritive tissue (endosperm) of the seed
- D: in the diploid nutritive tissue (perisperm) of the seed

A 70: Which storage organ does not exist in monocotyledons? (1 pt)

- A: onion
- C: tuber
- B: turnip
- D: rhizome

A 71: I - V lists plant species and 1 - 5 lists morphological and taxonomic characteristics. (2 pts)

I. onion (<i>Allium</i>)	1. monocotyledons
II. peach (<i>Prunus</i>)	2. dicotyledons
III. coconut palm (<i>Cocos</i>)	3. stone fruit
IV. oak (<i>Quercus</i>)	4. capsula
V. corn (<i>Zea</i>)	5. nut or caryopsis

Which assignments are correct?

- A: I: 1, 3 II: 1, 4 III: 1, 4 IV: 2, 4 V: 1, 5
- B: I: 1, 4 II: 2, 5 III: 1, 5 IV: 2, 3 V: 1, 3
- C: I: 1, 4 II: 2, 5 III: 1, 3 IV: 2, 5 V: 1, 3
- D: I: 1, 4 II: 2, 3 III: 1, 3 IV: 2, 5 V: 1, 5
- E: I: 1, 4 II: 2, 3 III: 1, 4 IV: 2, 5 V: 1, 3

MICROBIOLOGY / BIOTECHNOLOGY

A 76: A broth culture of *Escherichia coli* (50 ml inoculated with 0.5 ml (1 pt) of an overnight culture), grown at 37° C for 18 hours, contains:

- A: cells of *E. coli* which all divide at the same rate
- B: mostly endospores
- C: cells of *E. coli* which are all in the stationary growth phase
- D: cells of *E. coli* which are in all possible growth phases

A 77: Clear plaques in a bacterial lawn that do not increase in size are (1 pt) caused by

- A: the parasitic bacterium *Bdellovibrio bacteriovorus*.
- B: bacteriophages.
- C: bacteria producing antibiotics.
- D: bacteriocin-producing bacteria.

A 78: In a wastewater treatment plant flocks of activated sludge swim (2 pts) on the surface of the settling tank. Which physiological process causes this problem?

- A: nitrification
- C: denitrification
- B: sulphate reduction
- D: alcoholic fermentation

A 79: It is well known that heterologous DNA which is genetically stable (2 pts) can be established in a bacterial cell. Which of the following processes can be used to achieve this?

- I. conjugation
 - II. transformation of plasmid DNA
 - III. transformation of chromosomal DNA
 - IV. general transduction
 - V. special transduction
- A: I, III, V C: I, III, IV
B: I, II, III D: II, IV, V E: I, II, V

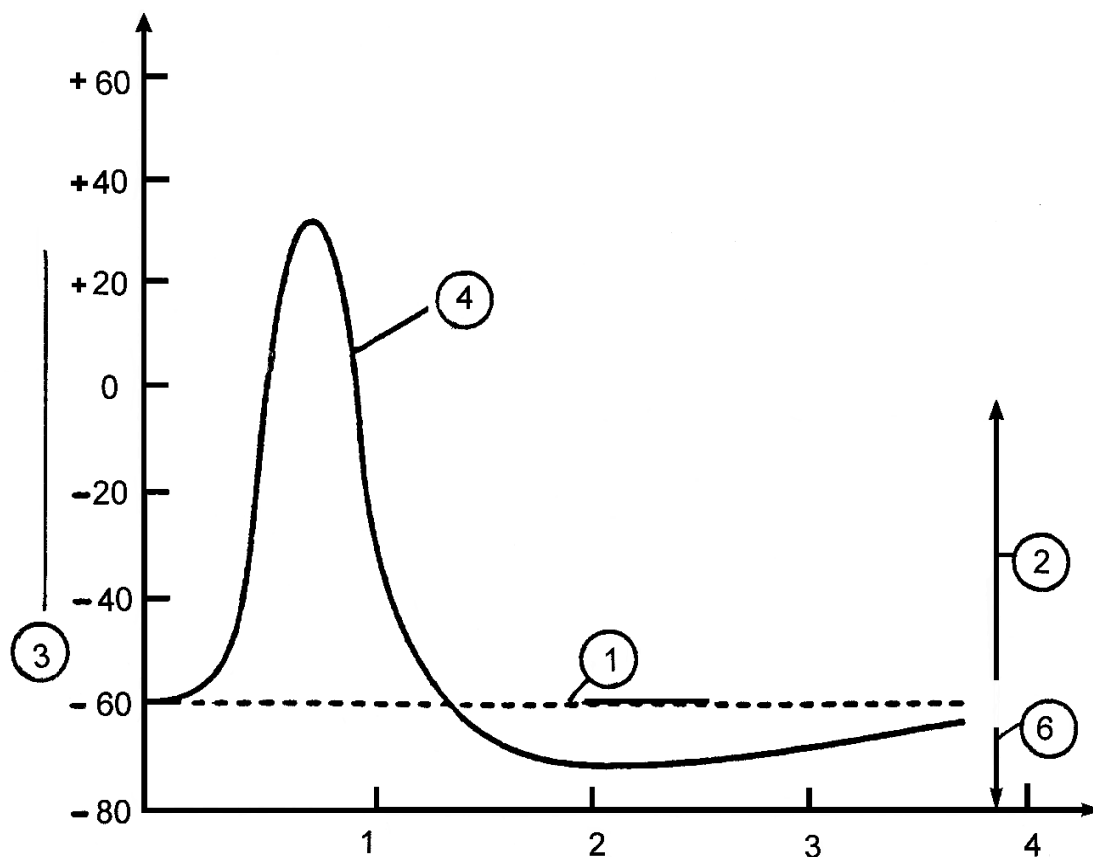
A 80: Which of the following organisms is used to transfer genes into (1 pt) higher plants?

- A: *Escherichia coli*
- C: *Agrobacterium tumefaciens*
- B: *Rhizobium trifolii*
- D: *Salmonella typhimurium*

5.2.2 Theoretical Test - Part B

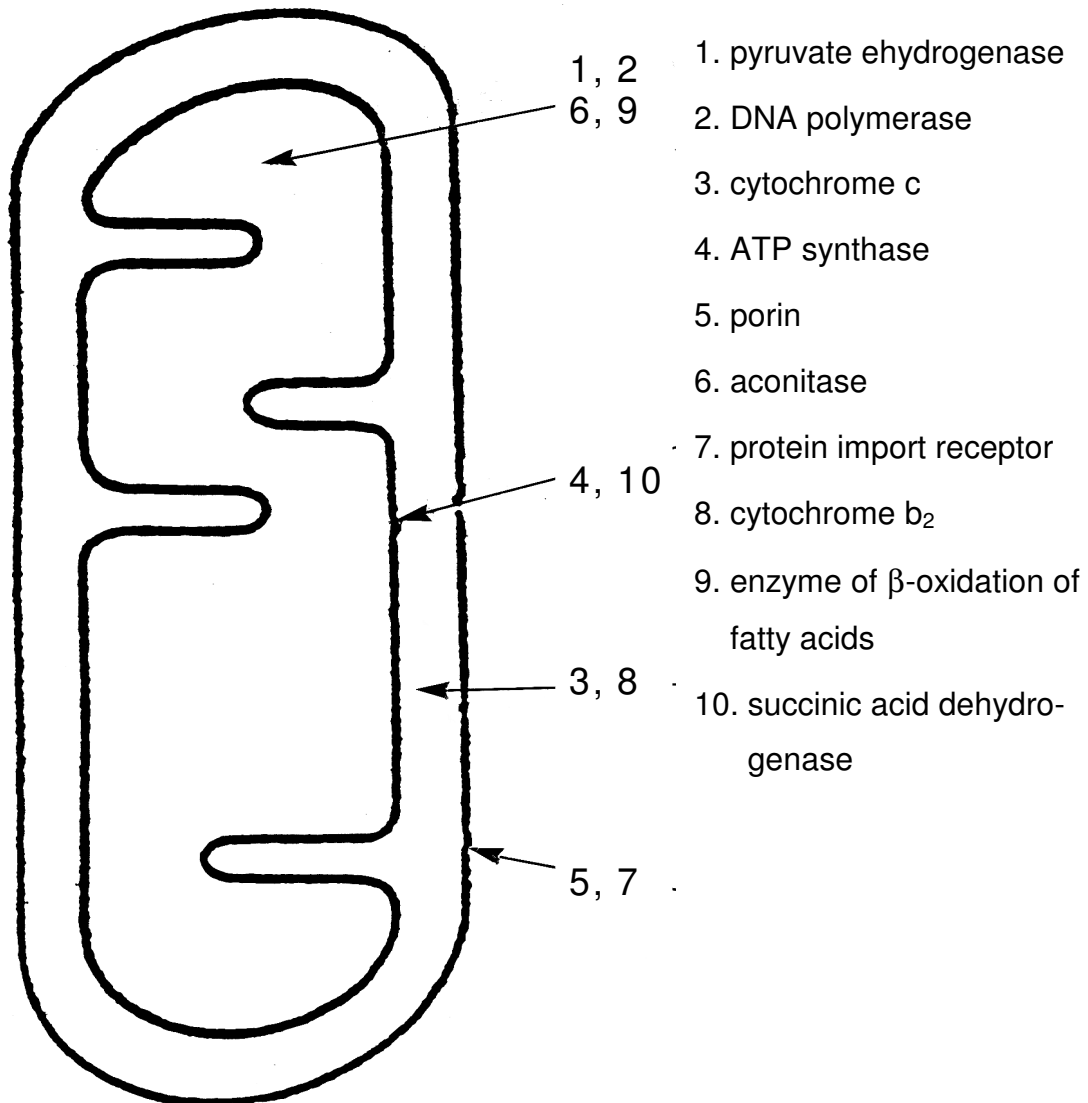
CELL BIOLOGY

B 1: Stimulating a giant axon of a squid at the time = 0 leads to a (1 pt) change in the membrane potential, which is measured with a microelectrode. The results are shown in the figure below. Label the figure using the adequate code numbers.

**Code numbers:**

1. resting potential
2. depolarisation
3. membrane potential (mV)
4. action potential
5. osmotic potential
6. hyperpolarisation

B 2: Match the molecules or proteins listed below with the various (2 pts) sub-compartments of mitochondria.



B 3: S K I P P E D .

(3 pts) **Bacterial ribosomes are distinguished from ribosomes coded by the nucleus because of their sedimentation (Svedberg-constant S) and their composition of rRNA. Name the ribosomes and their subunits according to the Svedberg constant and their rRNA composition. Fill in the correct code number in the corresponding space in the following table.**

Code numbers:

- | | | |
|--------------|-------------|-------------|
| 1. 5S rRNA | 3. 16S rRNA | 5. 23S rRNA |
| 2. 5.8S rRNA | 4. 18S rRNA | 6. 28S rRNA |

<u>bacterial ribosomes</u>	ribosomes coded by the nucleus
----------------------------	--------------------------------

.....70.....S	80.....S
---------------	---------------

<u>large subunit</u>	<u>large subunit</u>
----------------------	----------------------

.....50.....S	60.....S
---------------	---------------

<u>composition of rRNA</u>	<u>rRNA composition</u>
----------------------------	-------------------------

.....1.....	1.....
-------------	-------------

.....5.....	2.....
-------------	-------------

.....	1.....
-------	-------------

<u>small subunit</u>	<u>small subunit</u>
----------------------	----------------------

.....30...S	40.....S
-------------	---------------

<u>composition of rRNA</u>	<u>composition of rRNA</u>
----------------------------	----------------------------

.....3.....	4.....
-------------	-------------

.....	
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.....	
-------	-------

B 4: The average time of duplication of mammalian cells in cultures is (2 pts) 24 hours. Just divided cells growing on a cover-slip were treated with radioactively labelled (tritium) thymine for 10 minutes and immediately fixed. Figure I shows a typical microscopic section. Figure II shows the autoradiography with the dispersion of the silver grains of the section in figure I using the same magnification. Calculate the average duration of the S-phase in these cells with a maximum error of 30 minutes.

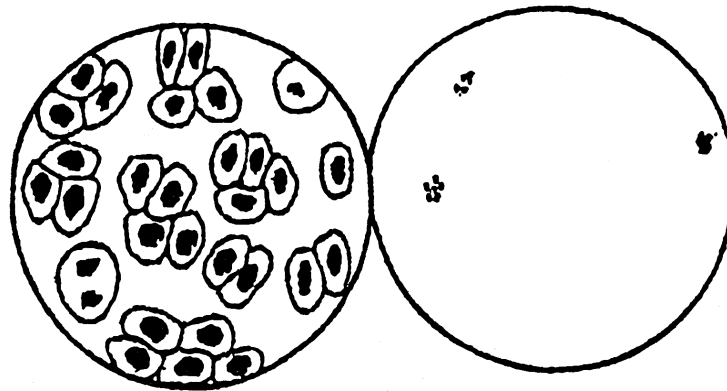


figure 1

figure 2

Average duration of S-phase:2,5 h.....

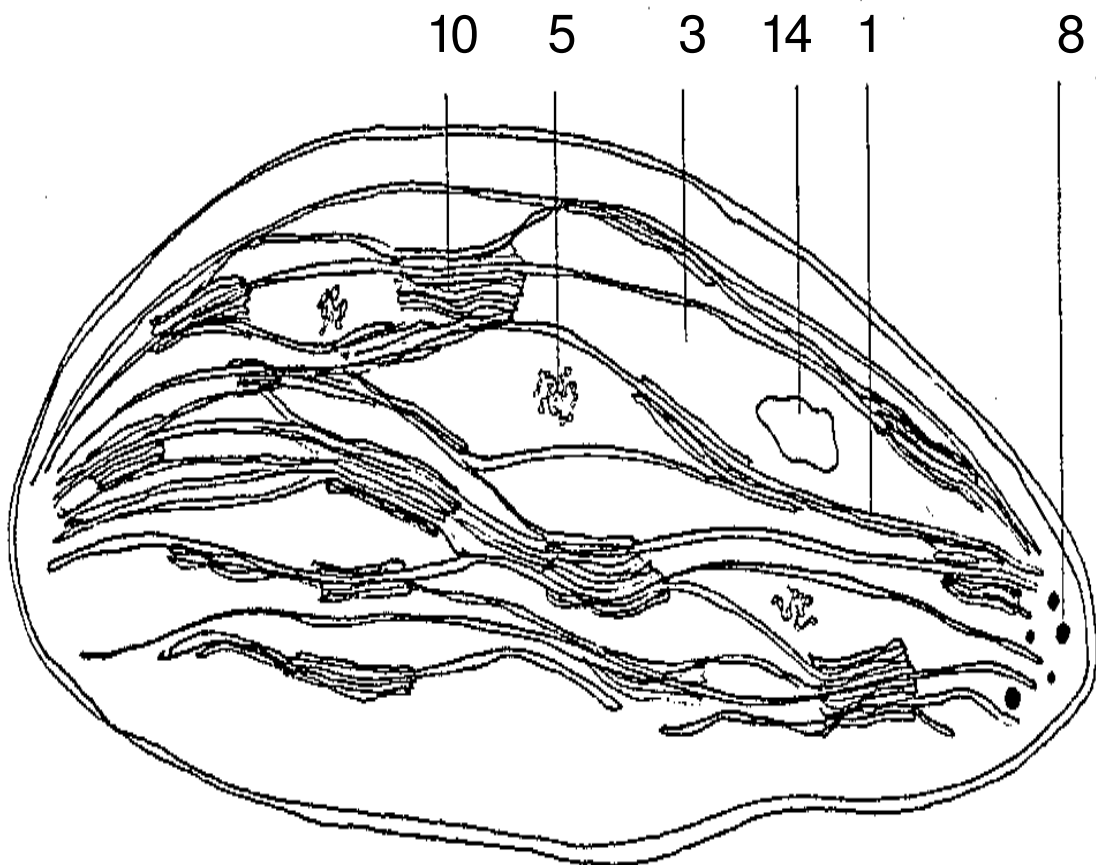
B 5: The following table shows structures of eukaryotic and prokaryotic cells as well as various membranes ('types of membrane'). Which cell structures possess which type of membrane ('types of membrane')? Mark the correct answer with a cross.

cell structures	presence or types of membranes		
	no membrane	monolayer membrane	double membrane
1. nucleus			X
2. lysosomes		X	
3. mitochondria			X
4. Golgi apparatus		X	
5. peroxisomes		X	
6. rough endoplasmic reticulum		X	
7. chloroplasts			X
8. vacuoles		X	
9. ribosomes	X		
10. centrioles	X		
11. nucleolus	X		
12. flagella of eukaryotes		X	
13. flagella of prokaryotes	X		

B 6: Label the marked structures of the chloroplast in the following (2pts) figure. Use the corresponding code numbers.

Code numbers:

- | | | | |
|---------------|-------------------|---|---------------------|
| 1. thylakoid | 5. nucleoid | 9. plastidome
(collection of plastids) | 13. pyrenoid |
| 2. cristae | 6. nucleolus | 10. granum | 14. grain of starch |
| 3. stroma | 7. amyloplast | 11. elaioplast | |
| 4. nucleosome | 8. lipid droplets | 12. prolamella body | |



B 7: The following table shows types of plastids and their specific (2 pts) characteristics. Match the characteristics to the types of plastids by marking with a cross.

Types of plastids	characteristics			
	starch formation	photosynthesis	coloured	contains DNA
Chloroplast	X	X	X	X
Chromoplast			X	X
Rhodoplast			X	X
Proplastid				X
Leuko-amyloplast	X			X

B 8: A cell produces and secretes a certain protein. Select from the (1 pt) list only those structures which are involved in these processes, to our present knowledge - and list the code numbers in the correct order of the processes.

1. peroxisome
2. free ribosome
3. Golgi apparatus
4. secretory vesicle
5. plasma membrane
6. vacuole
7. rough endoplasmic reticulum
8. lysosome

2, 7, 3, 4, 5

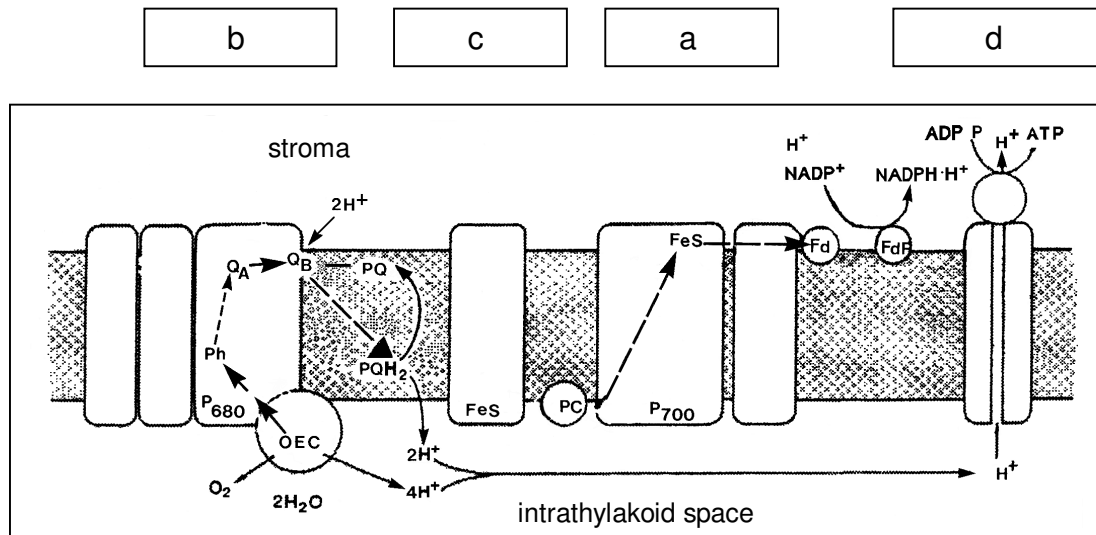
correct order of compartments:.....

ANATOMY AND PHYSIOLOGY OF PLANTS

B 9: The following figure shows a simplified schematic drawing of the most important photochemical complexes and their localisation in the thylakoid membrane of chloroplasts. Label the spots marked with with the four integral membrane protein complexes.

- a- photosystem I
- b- photosystem II
- c- cytochrome b_6 / f
- d- ATP synthase

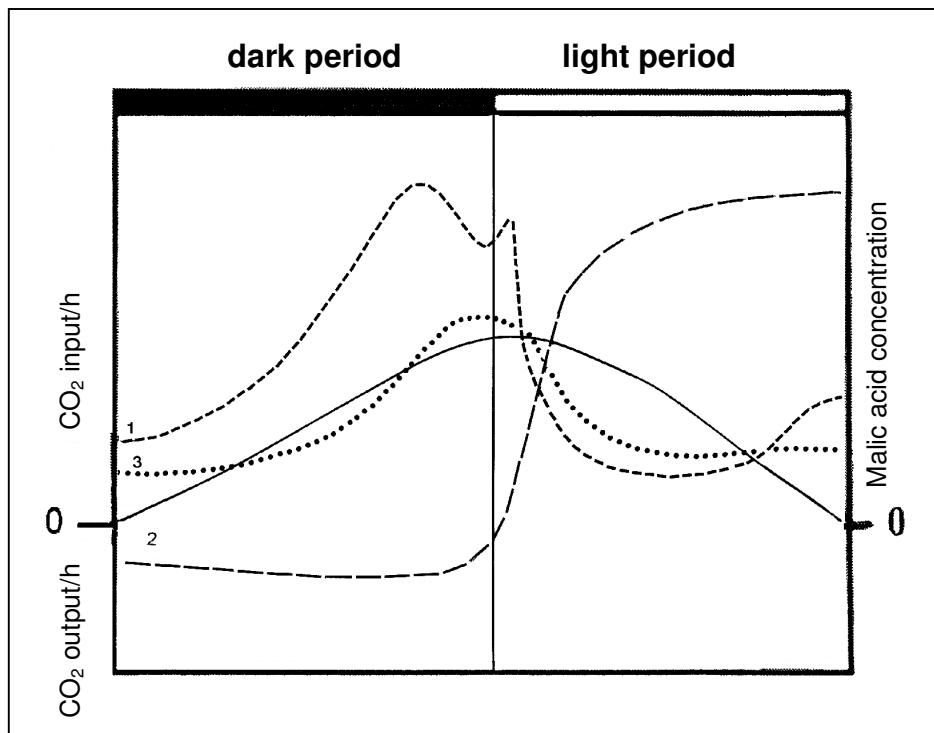
Mark with arrows (using a yellow pen) the electron flow from water as the e^- donator to NADP^+ as the e^- terminal acceptor.



legend: FdR = Ferredoxin-NADP⁺-Reductase; Fd = Ferredoxin; FeS = Iron-sulphur-centre; P₇₀₀, P₆₈₀ = chlorophylls of the reaction centres; PC = Plastocyanin; PQ = Plastoquinone; Q = Quinon; Ph = Pheophytine; OEC = water splitting complex .

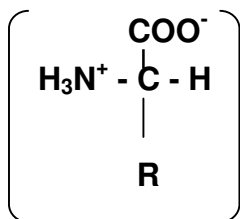
B 10: Plot with a yellow pen (1) the course of gas exchange of a typical CAM plant and with a pencil (2) the course of gas exchange of a C₃-plant using the figure.

The malic acid concentrations in the CAM plant vacuoles should be plotted in blue (3).



B 11: S K I P P E D .

(2 pts)



Amino acids can be divided in various classes according to the polarity of their R-groups. Classify the following amino acids due to the polarity of their R-groups by arranging the following code numbers in the table.

1: serine (Ser)

2: alanine (Ala)

3: valine (Val)

4: phenylalanine (Phe)

5: tyrosine (Tyr)

6: glutamine (Gln)

7: arginine (Arg)

8: glutamic acid (Glu)

9: aspartic acid (Asp)

classification of amino acids according to the polarity of their R-group (at pH 7)

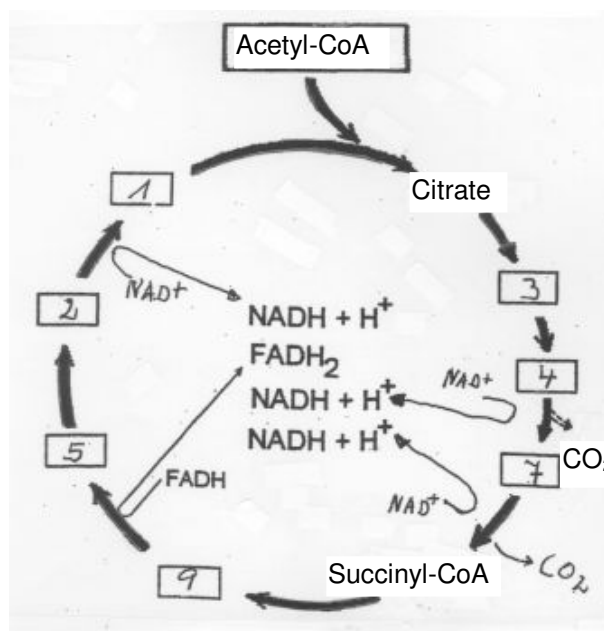
nonpolar (hydrophobic) R-groups	2, 3, 4
polar, but uncharged R-groups	1, 5, 6
negatively charged (acidic) R-groups	8, 9
positively charged (alkaline) R-groups	7

B 12: The figure below is a simplified scheme of the citric acid cycle.

(2 pts) Fill in the missing intermediates in the correct order by writing down the corresponding code number in the space provided.

- 1: oxaloacetate 4: isocitrate 7: 2-oxoglutarate
 2: malate 5: fumarate 8: malonate
 3: cis-aconitate 6: glyoxylate 9: succinate

Indicate by an arrow - like in the given example - the missing release of CO₂ as well as the three missing dehydrogenations in which the hydrogen is transferred to the NAD⁺.



B 13: Compare figure A and B. Indicate in the table, which structures in (2 pts) figure B correspond to those in figure A.

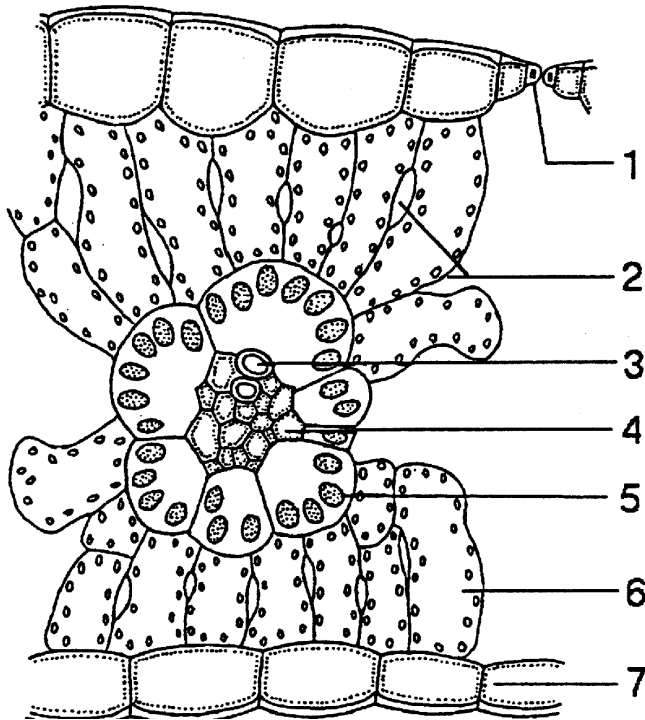


figure A	figure B
1	5
2	9
3	1
4	2
5	7
6	6
7	4

figure A

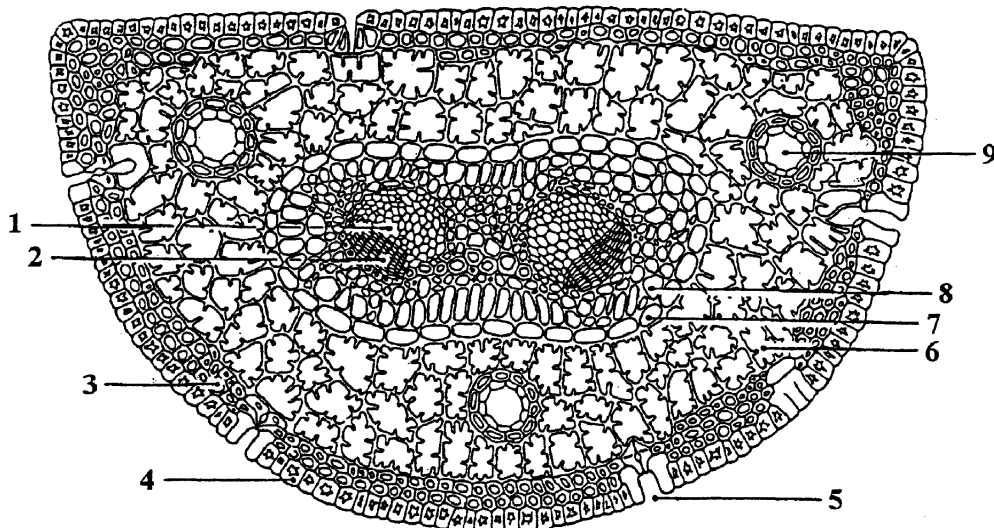
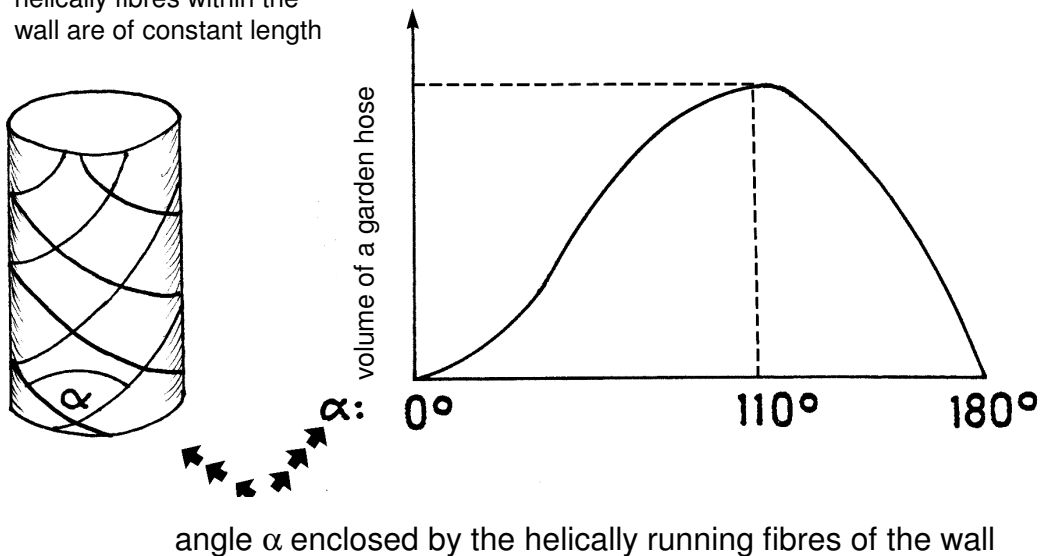


figure B

ANATOMY AND PHYSIOLOGY OF ANIMALS AND PLANTS

B 14: Within the wall of a garden hose, flexible fibres which can hardly be stretched run in crosswise helices enclosing an angle of $\alpha = 110^\circ$. Only at this angle, the water pressure within the hose cannot alter the length of the hose, its diameter, or its volume (see the maximum indicated in the curve presented).

volume of a cylinder, whose helically fibres within the wall are of constant length



angle α enclosed by the helically running fibres of the wall

In flat- and roundworms, flexible but inelastic fibres run crosswise helically within the body wall. The cross section of the body is flat in flatworms and circular in roundworms. The body volume is frequently changed by ingestion of food and defecation.

Therefore, which of the following statements is correct? Mark with a cross.

- ☐ $\alpha = 110^\circ$ in flatworms and $\alpha = 110^\circ$ in roundworms
- ☒ $\alpha = 110^\circ$ in flatworms and $\alpha = 150^\circ$ in roundworms
- ☐ $\alpha = 150^\circ$ in flatworms and $\alpha = 110^\circ$ in roundworms
- ☐ $\alpha = 150^\circ$ in flatworms and $\alpha = 150^\circ$ in roundworms

B 15: Microscopically small marine annelid worms differ in various aspects from their macroscopically large relatives. Mark the correct statement with a cross.
(1 pt)

- ☐ In the former, the relation of body surface to body volume is smaller than in the latter.
- ☐ In the former, the brain volume occupies a smaller part of the body volume than in the latter.
- ☐ A heart is often absent in the former, but always present in the latter.
- ☐ External fertilization is widespread among the former and does not occur among the latter.
- ☐ Planctic larvae are widespread among the former and do not occur among the latter.

B 16: The main waste product of protein and purine metabolism is urea in mammals and uric acid in birds. The solubility of these products in water is different. If mammals produced uric acid and birds urea (with an otherwise unchanged body organisation) why would this be disadvantageous to both? Mark the correct answer.
(2 pts)

- ☐ In both cases, the embryos would have substantial problems with excretion.
- ☐ In both cases, the adults would have substantial problems with excretion.
- ☐ The organs of mammals are unsuitable for excreting uric acid.
- ☐ The organs of birds are unsuitable for excreting urea.
- ☐ In the case of urea formation, birds would lack important raw materials for feather formation.
- ☐ In the case of uric acid formation, mammals would lack important raw materials for hair formation.

B 17: Which one of the following statements related to the neuronal (2 pts) basis of learning is correct?

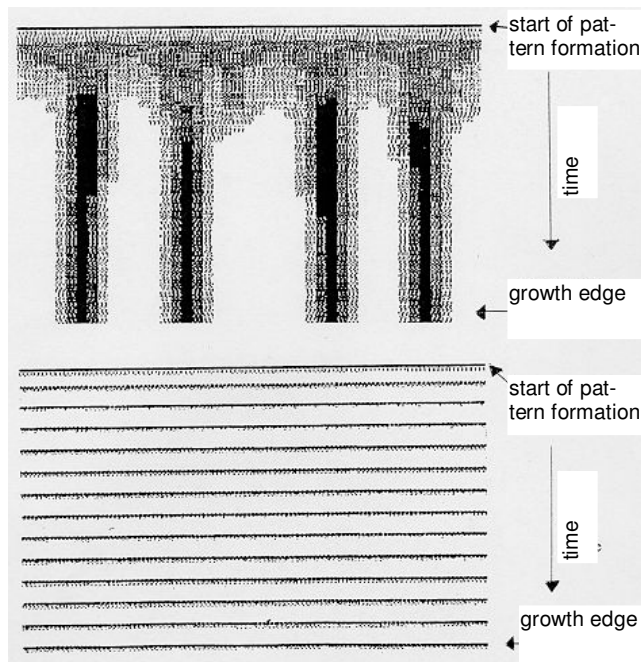
- ☐ Learning immediately induces the formation of specific memory molecules.
- ☐ Learning immediately induces the formation of specific memory synapses.
- ☐ Learning immediately induces the formation of specific memory neurons.
- ☐ Permanently, too many memory molecules are formed. Learning improves the fitness of a few of them which survive, while the less fit ones are decomposed.
- ☐ Too many synapses are formed constantly. Learning improves the fitness of a few of them which survive, while the less fit ones are decomposed.
- ☐ Too many neurons are formed constantly. Learning improves the fitness of a few of them which survive, while the less fit ones are broken down.
- ☐ Selection due to differences in fitness does not occur in the brain. Neurons are activated by learning and otherwise rest.

B 18: S K I P P E D .

(3 pts) Inspired by nature, the presented patterns I and II were generated by a computer. Pattern formation takes place only at the growth edge (at the bottom in the figures). An activator stimulates pigmentation (responsible for black parts of the patterns), while this process is inhibited by an inhibitor (responsible for white parts of the patterns). Patterns form, because the concentrations of activator and inhibitor vary in time and space. The activator promotes its own production by autocatalysis and, additionally, promotes production of the inhibitor by catalysis. Both activator and inhibitor can diffuse and can be broken down. The example shows, the laws of pattern formation differ only by their rates of diffusion D and the rate of decomposition R of activator and inhibitor.

In order of increasing size the values are (A stands for activator, I for inhibitor):

$$\begin{array}{ll} D_A = 0.005 \text{ and } 0.1 & D_I = 0.4 \text{ in both cases} \\ R_A = 0.01 \text{ and } 0.06 & R_I = 0.003 \text{ and } 0.015 \end{array}$$



Write down which values correspond to pattern I and which to pattern II.

- values of pattern I: **0,005** $D_A = 0,005$ $R_A = 0,01$ $R_I = 0,015$
- values of pattern II: **0,1** $D_A = 0,1$ $R_A = 0,06$ $R_I = 0,003$

ETHOLOGY

B 19: Sexual selection of swallows (*Hirundo rustica*)

(4 pts) Reproduction and parental care cause costs and yield benefits.

The right mate selection can optimize the cost-benefit-ratio (e.g. measured by the number of eggs per nest or by the number of the hatched and fledgling offspring). Male and female swallows differ in their behaviour and in the length of their outer tail feathers (see table below).

Characteristics of swallows

swallow	Body length	weight	feathers other than mil feathers	tail feather	arrival from Africa
male	19 cm	18-26 g	identical E E to females	106 mm (100%)	before E E (females)
female	19 cm	18-26 g	identical Γ Γ to males	91 mm (84% of 106 mm)	after Γ Γ (males)

Courting male swallows change their flight path in the presence of a female and show two long feathers in the spread tail fan. These feathers might indicate certain features of males as a secondary sexual characteristic. If that is correct, experiments with modified ordinary tail feather lengths should yield various results. In the picture males are shown with ordinary and experimentally modified tail feathers.



The next table shows the results of three groups of males (A, B, C), which had different feather lengths.

Results of an ethological experiment

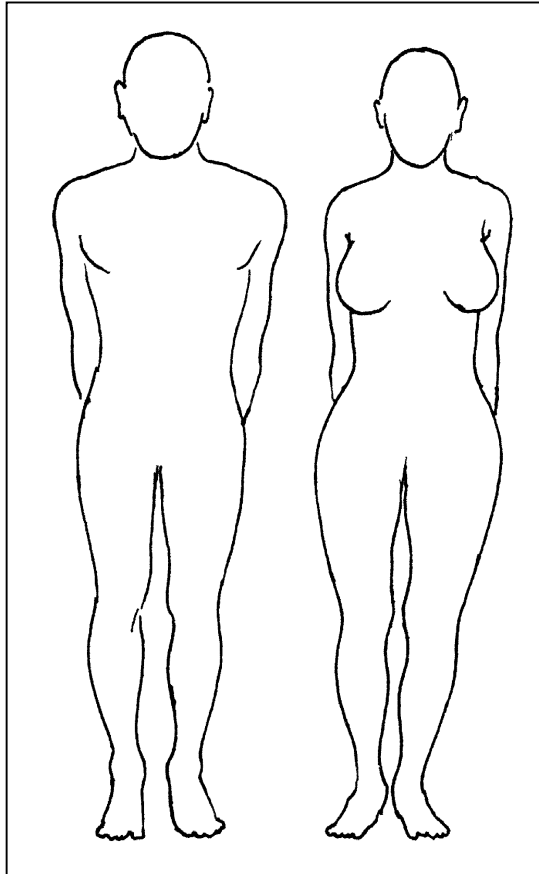
male tail feathers	A: 106 mm (ordinary)	B: 85 mm (shortened)	C: 127 mm (lengthened)
time of courting	6.8 days	12.3 days	3.4 days
offspring / season	5.0 young	3.3 young	8.4 young
aggression $\Gamma \Gamma$ (males) against $\Gamma \Gamma$ (males)	as B, C	as A, C	as A, B

Analyse these results and answer the following questions:

- (1) What is the time-saving of a pair (in %) with the duration of incubation of 14 to 16 days, if the female does not mate with male A but with male C?
.....21,3.....to24,3..... %
(incubation of 14 to 16 days)
- (2) How many offspring on average does a female have with a male C (long tail feathers) compared to an ordinary male A (result in %)?68..... %
- (3) According to the results, what determines the most successful reproduction of breeding pairs? Mark with a cross.
 - ☐ the condition and the behaviour of females in the choosing of a mate against males (female choice)
 - ☐ the condition and the successful competition of males against each other (male-male competition)
 - ☐ the condition and the successful competition of females against each other (female-female competition)
 - ☐ the condition and the behaviour of males in the choosing of a mate against females (male choice)

B 20: Preference of male and female body proportions in selective (4 pts) model experiments.

In a very extensive investigation (n = 10,000 people) children, youths and adults were each shown two outline pictures of the male and female body (models).



After the comparison each person stated their preferred model. The following table (next page) shows the results for 4 - 20- year-olds, divided by

- I. age of the experimental subject
- II. sex of the experimental subject
- III. sex of the preferred model

Selective behaviour of males and females (body outline in double selection)

age (years)	E choice by (males) Γ Γ %	Γ choice by (males) Γ Γ %	Γ choice by (females) E E %	E choice by (females) E E %
4	45	55	43	57
5	43	57	44	56
6	39	61	44	56
7	35	65	42	58
8	33	67	39	61
9	32	68	37	63
10	28	72	36	64
11	26	74	36	64
12	26	74	37	63
13	28	72	42	58
14	34	65	50	50
15	42	57	57	43
16	50	50	63	37
17	57	43	66	34
18	63	37	67	33
19	67	33	63	34
20	68	32	58	42

Evaluate the table and mark the correct answer with a cross.

(1) Which models do younger females prefer?

☐

male models

☐

female models

(2) Which models do older males prefer?

☐

male models

☐

female models

(3) Between which ages do females change from non-preference (< 50%) to preference (> 50%) of male models?

☐

at 14

☐

at 15

☐

at 16

p.t.o.

- (4) Between which ages do males change from no-preference (< 50%) to preference (> 50%) of female models?

- ☐ at 14
☐ at 15
☐ at 16

- (5) Which hypotheses correspond to the data?

Mark the correct statements with a cross.

- ☐ **Hypothesis 1:** Secretion of growth hormones from the tissue of the thyroid gland causes the orientation for mates of the opposite sex. The hormonal effect starts at a different time in boys and in girls.
- ☐ **Hypothesis 2:** Sexual learning optimises the preference of the opposite sex. The hormonal effect starts at a different time in boys and in girls.
- ☐ **Hypothesis 3:** The neurophysiological maturation of the preference concerning the opposite sex is promoted by the increasing amount of oestrogen (E E) and testosterone (Γ Γ) during puberty. It develops at different times during maturation in boys and girls.

B 21: Mate selection in grasshoppers (*Tettigoniidae*, crickets)

- (2 pts) In the population of one species A males which are ready to reproduce sing on fixed locations. The females are orientated towards the songs, approach the males and get on them (inversion of the sexual role compared to other grasshoppers). The males produce one spermatophore during their lifetime, and it can only be given to one female. Nevertheless, a female can produce more than one clutch of eggs and mate with several males in succession. Contrary to the otherwise rare female reproduction products (compared to the number of male gametes) in this case the females should have a measurably higher success rate in reproduction. An investigation showed the following results:

	average weight of females before eggs are laid	average number of eggs per female
Rejected by males	3.23 g	30
Accepted by males	3.71 g	48

Answer the following questions according to the results:

(1) What is the average weight of a female without eggs?

- ☐ 2.43 g
☐ 2.24 g
☐ 2.32 g

(2) How many eggs can each of four males fertilize, when it mates with a female that produces four clutches of eggs in succession and weighs 3.87 g, 3.74 g, 3.52 g and 3.74 g, respectively when copulating?

male 1 with female 154..... eggs

male 2 with female 149..... eggs

male 3 with female 141..... eggs

male 4 with female 149..... eggs

B 22: Spectrum of prey and food selection of the oyster catcher

(2 pts) (*Haematopus ostralegus*)

Oyster catchers are birds that look for prey in the shallow water of sea coasts, e.g. bivalve. The shells in a habitat are of different sizes and these sizes correlate with the amount of energy in the shell meat:

- Small shells can be opened fast and easily (costs) but provide less food energy (benefit).
- Big shells provide more energy (benefit), but they are more difficult and time-consuming to open (costs).

The following table shows

- I. the total amount of energy of big shells of different size
- II. the frequency of distribution of all shells present
- III. the frequency of distribution of the shells selected

Number and energy content (relative units) in shells living in a feeding territory that were selected by birds

total amount of energy per shell (E)	shells present (n)	energy present (E)	shells selected (n)	energy consumed (E)
1	1	1	0	0
2	17	34	0	0
3	11	33	1	3
4	9	36	3	12
5	6	30	6	30
6	7	42	7	42
7	20	140	11	77
8	22	176	10	80
9	19	171	14	126
10	13	130	13	130
11	6	66	6	66
12	2	24	2	24
13	1	13	1	13
1 to 13	134	sum 896	74	sum 603

Analyse and evaluate the table and answer the following questions:

- (1) What total amount of energy could be expected from 74 shells that were randomly selected from 134 shells?
.....495..... energy units E (proportional)
- (2) What is the absolute total amount of energy and the total amount of energy in percentage, that can additionally be consumed from 74 shells that were selected by birds.

.....108..... energy units R (real)

.....21,8..... % benefit

GENETICS/EVOLUTION

B 23: Which characteristics does an interspecific hybrid from crossing (2 pts) **diploid flowering plants have? From the list below, select and write down the code numbers of the correct reasons or explanations in the circles below.**

properties of the hybrid

- ☒ 3 amphihaploid (haploid form of amphihaploid)
- ☐ amphidiploid (allopolyploid with double the normal chromosome number)
- ☐ fertile, but its offspring are sterile
- ☐ usually fertile, if the parents belong to the same genus
- ☐ sterile and therefore not suited for plant breeding
- ☒ 6 It can be made fertile by the toxin colchicin.
- ☐ It can be made fertile by the phytohormone β -indolelactic acid (IAA).

Code numbers for reasons and definitions:

1. The interspecific infertility is a significant feature of the species.
2. The embryonic development is impaired.
3. Each type of chromosomes is present only once.
4. Two chromosome sets are present.
5. The formation of fertile gametes is impaired.
6. The treatment facilitates the meiotic chromosome pairing to take place.
7. The treatment facilitates fruit maturation.
8. The treatment does not lead to the mentioned result.

B 24: In DNA replication the 3'-end priming problem is known (end replication problem). From the list below, write in the circles in the corresponding genophores (chromosomes DNA) the numbers of the phenomena by which the problem is solved or avoided.

genophores

- main genophores of eubacteria
- plasmids of eubacteria
- main genophores of archaea bacteria
- mitochondrial DNA
- plastid DNA
- chromosomes of nuclei
- DNA of lambda-phages
- DNA of adenovirus
- DNA of retrovirus

code numbers for phenomena:

- 1. circular nature of DNA
- 2. pre-replicative circularization of the DNA
- 3. replication jumps from the 5'-end of the leading, to the 3'-end of the lagging strand
- 4. telomerase activity
- 5. specific primer-independent DNA replicase
- 6. transcription / reverse transcription
- 7. polyadenylation

B 25: Isolated protoplasts of two albino mutants of potatoes were fused. A variegated plant was obtained. Where are the mutant genes, which have caused the chlorophyll deficiency, located? A new mutation, plastid communication and plastid DNA recombination should be excluded.

Mark the correct answer with a cross.

- ☒ in the nucleus of one mutant, in the plastids of the other one
- ☐ in the nucleus of both mutants
- ☐ in the plastids of both mutants

B 26: Compare the following types of anaphase.

(2 pts) **Which of the following statements are correct for anaphase I of meiosis? Which statements are correct for the anaphase of mitosis? Write the correct code numbers from the list below, in the spaces beside the anaphase types. Pay attention to the fact that chromatides are subunits of chromosomes.**

1. phase I of meiosis:1, 3, 4, 5.....

2. anaphase of mitosis: ..1.....5.....

code numbers:

1. Chromosomes move to the poles.

2. Single chromatids move to the poles.

3. The moving chromosomes consist of two chromatides.

4. The chromosomes can carry two alleles of a gene.

5. One haploid set of chromosomes can move to each pole.

6. There are four spindle poles per cell.

B 27: In the course of the evolution of plants the haploid life phase, i.e. the gametophyte, has been more and more reduced and the diploid life phase, i.e. the sporophyte, has become the dominant generation. Write the number of the following taxa in the correct order according to the increasing dominance of the diploid life phase.

I. *Ulothrix (Chlorophyta)*

II. *Chlorella (Chlorophyta)*

III. *Pinus (Pinophyta)*

IV. *Polytrichum (Bryophyta)*

V. *Rosa (Magnoliophyta)*

VI. *Navicula (Phaeophyta, Diatomeae)*

..... II, I, IV, III, V, VI.....

B 28: S K I P P E D .

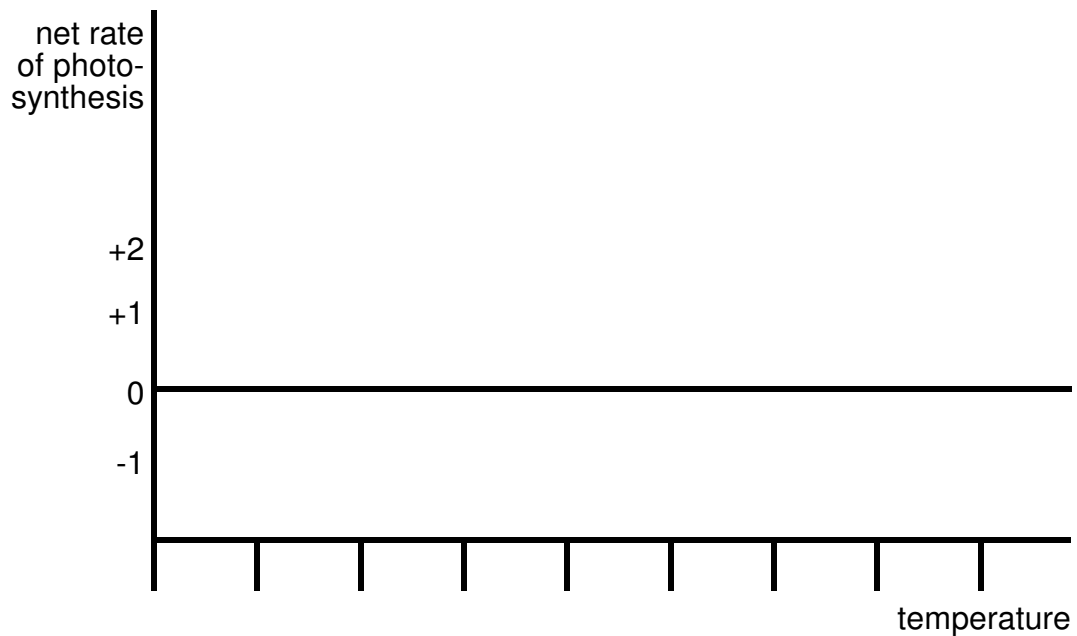
(1 pt) **The principle of natural selection is essentially important to Darwin's theory of natural selection, and this is essentially important to our understanding of evolution. According to the principle of natural selection, exponential growth of populations forces selection to occur again and again.**

Which of the following statements is correct? Mark with a cross.

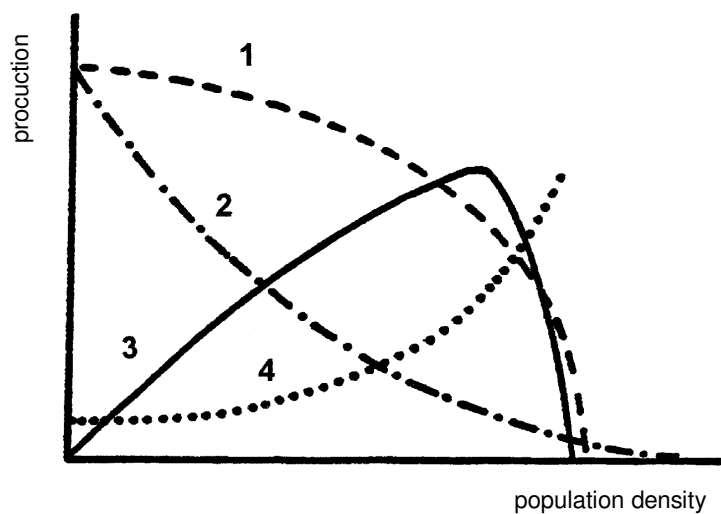
- ☐ The principle of natural selection is a rule, as it dominates the evolution of many, but not all species of organisms.
- ☐ The principle of natural selection is a law which dominates evolution of all species of organisms only.
- ☐ The principle of natural selection is a natural law which is only valid within the biotic world where it dominates the phylogenesis of all species and the ontogenesis of all organisms.
- ☐ The principle of natural selection is a natural law which in both the biotic and abiotic world dominates all processes in which self-replication of basic entries occurs.
- ☐ The principle of natural selection is only seemingly valid in evolution. In reality, it dominates nothing else than self-organization of dynamic order in abiotic systems.
- ☐ The principle of natural selection is not a natural law but, rather, a tautology: The fittest are the most successful in survival and reproduction and, hence, they are the fittest.

ECOLOGY

- B 29:** The dependence of the net photosynthesis on temperature follows a mathematical function. Show the dependence of the net photosynthesis on temperature using the axes below, of a C_3 plant at light saturation.



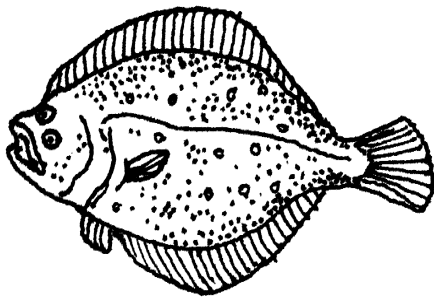
- B 30:** The following graph shows the changes in values of various biological factors/parameters in relation to increasing density of fish.



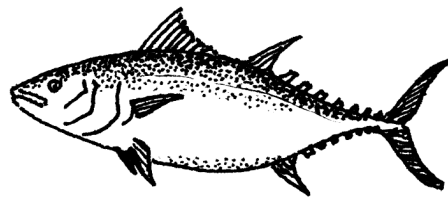
**Match the curves 1, 2, 3 and 4 from the graph to the terms in the table.
Match each curve only once.**

	curve from the graph
total fish production	3
individual growth	2
food potential not used	1
expenditure of energy of each fish in the search for food	4

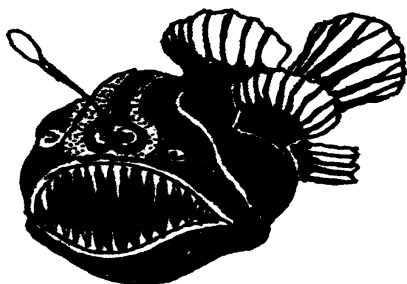
**B 31: The following pictures show fish from various locations.
(2 pts)**



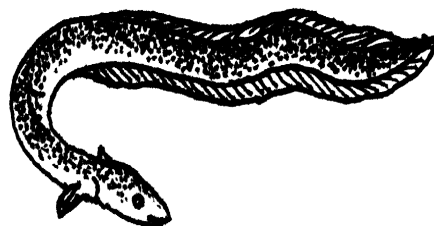
A. flat fish
at the bottom of shallow sea



B: tuna
in the open water in sea



C: angler-fish
at the bottom of deep sea



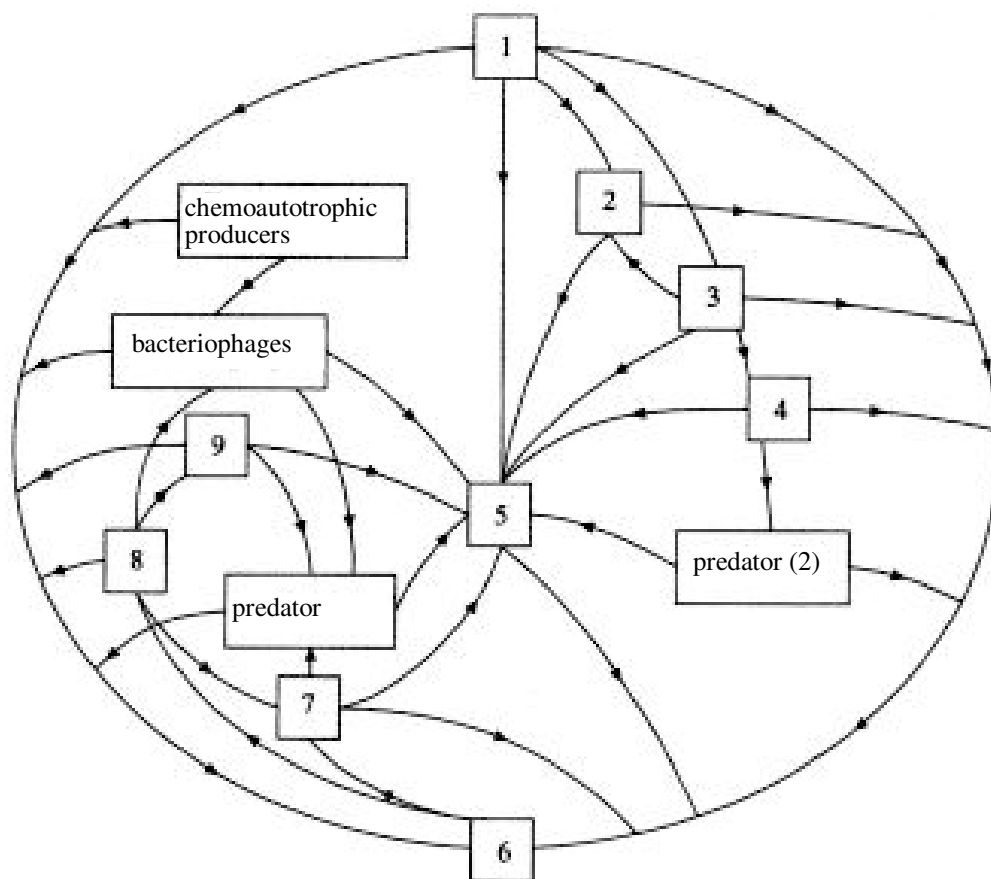
D: eel
predominantly at the
bottom of fresh water

Match the fish A, B, C and D from the picture to the gill surfaces in the table.

Gill surface per gramm body weight (in arbitrary units) of fish of various locations.

gill surfaces	fish
2551	B
902	D
462	A
51	C

B 32: The following figure shows a section of a food web in an ecosystem. (3 pts) **tem.**



Write down the numbers from the figure at the appropriate place in the following table.

Parts from a food net in an ecosystem

part from the food web	assigned number from the picture
organic waste	6
pantophage (omnivore)	2
saprophage (delritivores)	7
photoautotrophic organisms (producer)	1
phytophage (herbivore)	3
decomposers	8
mycophages	9
parasitic animals	5
zoophage (predator I)	4

B 33: Genetically identical seedlings of the sunflower (*Helianthus, Asteraceae*) were planted in two neighbouring gardens A and B. After three months the plant height was measured with the following results:

garden	number of plants	average height (in mm)	standard deviation
A	12	1.6	0.3
B	16	1.4	0.2

How can the differences be evaluated regarding the growth?

- (1) The differences are accidental. The growth conditions in both gardens are the same (null hypothesis).
- (2) The differences do not occur by coincidence. The growth conditions in garden A and B are significantly different (null hypothesis is rejected). You have to check the question with the help of the t-test.

The following t-values should be used.

probability	critical t-value
90 %	1.32
95 %	1.71
97 %	2.06
99 %	2.48
99.5 %	2.78

Exercises:

- (1) Calculate the standard deviation of each difference between the average values for garden A and B for this t-test. Use the following formula:

$$S = \sqrt{\left\{ \left(\frac{S_A^2}{n_A} \right) + \left(\frac{S_B^2}{n_B} \right) \right\}}$$

Result: S = 0,10

- (2) Calculate the t-value and determine by which probability the null hypothesis can be rejected (the differences are significant). Mark with a cross.

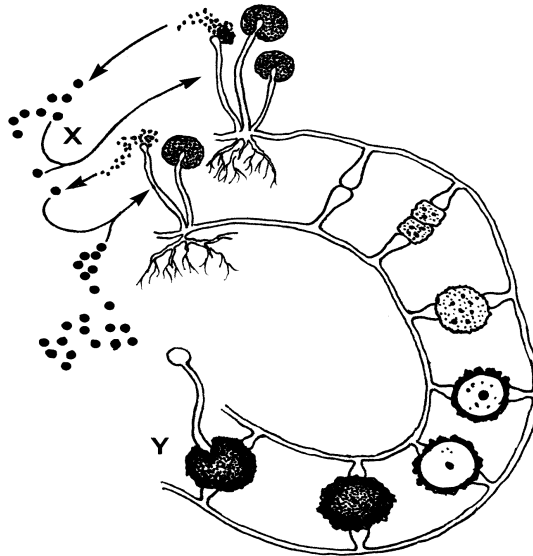
probability ☐ < 90 % ☒ 95 to 97.5 % ☐ 99 to 99.5 %

☐ 90 to 95 % ☐ 97.5 to 99.0 % ☐ > 99 %

SYSTEMATICS/TAXONOMY

B 34: The following illustration shows the life cycle of a fungus.

(2 pts)



(1) Which fungus group does the fungus whose life cycle is shown belong to? Mark the correct answer with a cross.

- ☐ *Ascomycota*
☐ *Basidiomycota*
☒ *Zygomycota*

(2) Which kind of spores are spread at X?

- ☐ haploid meiotic spores
☒ haploid mitotic spores
☐ diploid spores

(3) Which way of life is typical for this fungus?

- ☒ This fungus is a saprophyte.
☐ This fungus is a parasite.

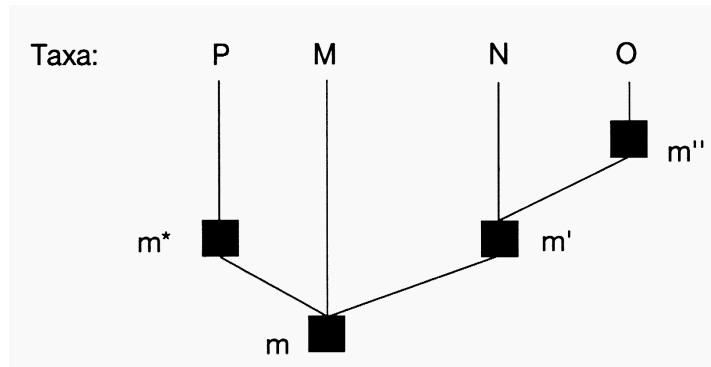
(4) Where does the meiosis take place?

- ☐ The meiosis takes place at X.
☐

X The meiosis takes place at Y.

B 35: A synapomorphy is a homologous characteristic of *various species* (therefore *Syn-*), that has evolved/is derived (apomorphic) compared with the primary absence or previous state of that characteristic in other species. The original characteristic m evolves into m' and m'' on the one hand and to m^* on the other (see figure).

Note that m' is apomorphic compared with m , but not compared with m'' . M , N , O and P could be the taxa in which the characteristics occur.



Mark the statements which are compatible with the definition of synapomorphy.

- ☐ m' is a synapomorphy of N compared with m in M
- ☐ m' is a synapomorphy of N
- ☐ m' is a synapomorphy of NO (N plus O) compared with m in M
- ☐ m'' is a synapomorphy of O compared with m in M
- ☐ m'' is a synapomorphy of O
- ☐ m'' is a synapomorphy of O compared with m^* in P
- ☐ m'' is a synapomorphy of O compared with the primary absence of m'' in P
- ☐ m'' is a synapomorphy of N compared with the primary absence of m' in M

- ☐ the secondary lack of m' in O is a synapomorphy of O compared with m' in N

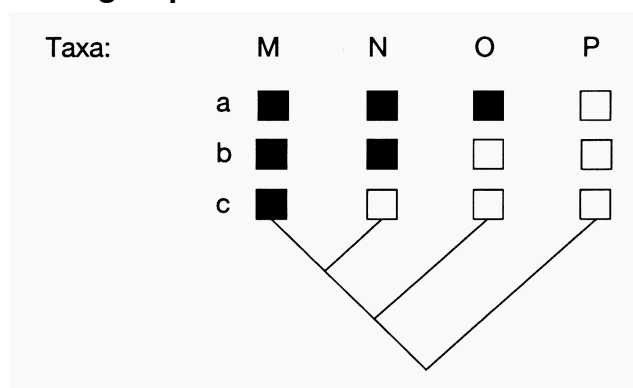
B 36: A set of species is called monophyletic if it contains exactly all known species which are descendants from one well characterized

but nevertheless hypothetical stem species.

For calling a studied set of species monophyletic the proof of at least one synapomorphy of this set is

- ☐ necessary and sufficient
- ☒ necessary, but not sufficient
- ☐ not necessary, but sufficient
- ☐ not necessary and not sufficient

B 37: In cladistics (a method of systematics) the outgroup comparison (2 pts) plays an important role. It works as follows: The states of the characteristics a, b and c marked black and white, are present in the taxa M, N, O as shown in the figure. Suppose MNO (M plus N plus O) are monophyletic. P is then its outgroup. As the states marked black occur only within MNO they are apomorphic compared with the states marked white which occur both in MNO and the outgroup P.



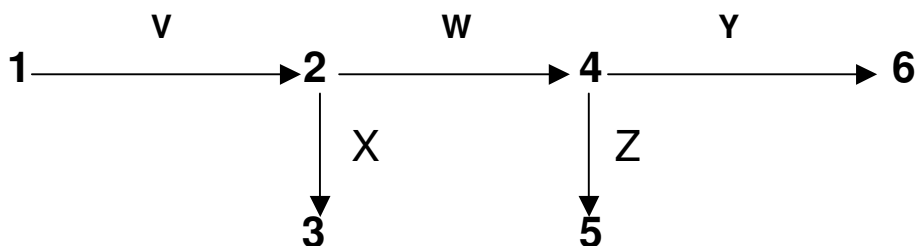
What is correct? Mark with a cross.

- ☐ 1. The states of the characteristics m, n and o marked black are not only apomorphies but even synapomorphies of MNO, MN and M compared with the states marked white.

- ☐ 2. Because of the evidence of synapomorphy, the states of m, n and o marked black are even homologies of MNO, MN and M, respectively.
- ☐ 3. If the states of a, b and c marked black do not occur outside of MNO, MN and M, respectively, these sets of species are monophyletic.
- ☐ 4. The outgroup comparison method has led to a circular argument.

MICROBIOLOGY/BIOTECHNOLOGY

B 38: The diagram represents a sequence of reactions taking place in a (2 pts.) bacterium, in which amino acids are produced from one another by the action of specific enzymes. Numbers 1 to 6 represent different amino acids; letter V to Z represent different enzymes. All the amino acids are essential for survival. The original strain of the bacterium required only amino acid 1. A mutant strain of the bacterium could only survive when provided with amino acids 1,2 and 5 in its culture medium. Which enzymes were missing in the mutant strain?



In the mutant the following enzymes were missing: V, Z

B 39: Mark the correct answers with a cross. (2 pts)

- ☐ 1. Yeast cells used for wine production tolerate up to 14 % ethanol.
- ☐ 2. Citric acid is produced industrially with the help of *Aspergillus niger* in an anaerobic process.
- ☐ 3. By mutagenesis and strain selection bacterial and fungal production of antibiotics is increased on a higher level than by gene amplification.
- ☐ 4. Cephalosporins are produced by bacteria.
- ☐ 5. The flock of activated sewage sludge is the active microbiological aggregate in anaerobic waste water treatment.
- ☐ 6. It was found in practice that microorganisms modified by gene technology for in situ remediation of organic contaminations in the soil or ground water are inferior to the adapted, natural microbial community with regard to their cleaning performance.

B 40: What is correct? Mark with a cross. (1 pt)

- ☒ H₂ can be used as an electron donor by chemolithoautotrophic bacteria.
- ☐ Sulphate reducing bacteria are inferior to methanogenics bacteria when competing for hydrogen.
- ☒ H₂ is used for the denitrification of drinking water.

5.2 Practical Tests

5.2.1 Laboratory 1: Systematically-anatomically orientated

Introducing remarks:

In laboratory 1 you have to solve both of the following tasks:

Task 1: Anatomical examination of a plant leaf

Task 2: Examination and systematic classification of four different plant species.

Read the questions for both tasks and then decide in which order you would like to work.

Please, do not forget to write down your personal data (Surname, First name, Country, Code Number) in the space provided.

Task 1: Examination of a plant leaf

(14 pts)

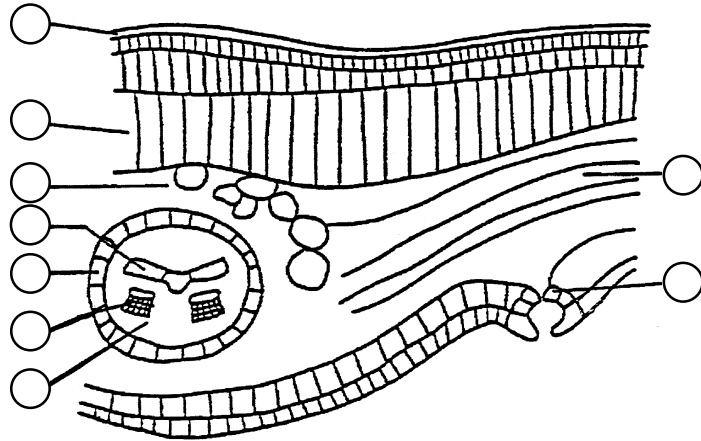
Introducing remarks:

The required equipment and chemicals for the examination are at your bench. Use the equipment and chemicals suitable for your examinations. Make several cross sections through the petiole of the leaf (figure 1). Stain these cross sections with the aqueous solutions of the dyes Astra-blue (stains cellulose) and Auramin (stains lignin) and analyse the anatomic structures under the microscope. As supplementation you can make longitudinal and surface sections.

1 (1.1) Label the given schematic drawing (figure 1). Write down the code numbers of the given anatomical structures next to the correct reference lines.
(9 pts)

Code numbers:

1. Cuticula
2. Epidermis
3. Hypodermis
4. Spongy parenchyma
5. Palisade parenchyma
6. Vascular bundle sheath
7. Xylem with lignin
8. Phloem with sieve cells and parenchyma
9. Transfusion tissue
10. Accessory transfusion tissue
11. Stomatic cells



1 (1.2) Which of the following systematical groups does the object refer to? Mark with a cross.
(2 pts)

- ☐ Fern (*Polypodiophyta*, *Pteridophyta*)
- ☐ Gymnosperms
- ☐ Angiosperms

1 (1.3) Evaluate the object from the ecological point of view.
(3 pts) **Mark with a cross.**

- ☐ Hydrophyte (swamp or water plant)
- ☐ Hygrophyte (plant with adaption to damp locations)
- ☐ Mesophyte (plant with adaption to balanced locations)
- ☐ Xerophyte (plant with adaption to dry locations)

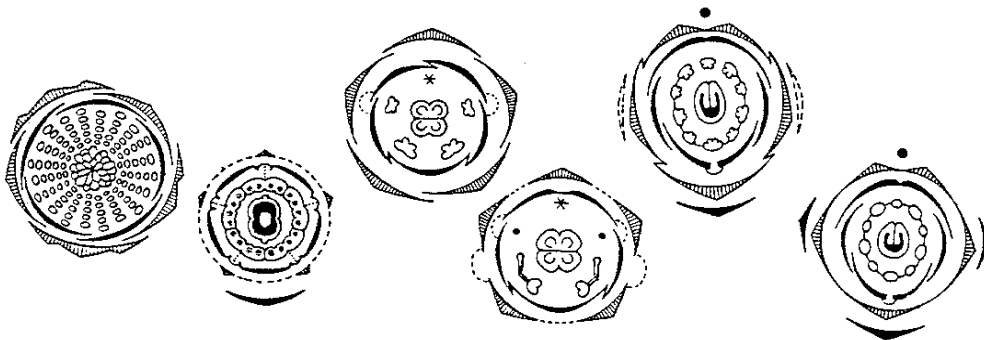
Task 2: Examination and systematic classification of four different (24 pts) plant species.

Introducing remarks:

On your bench there are four vessels. In each vessel there is a flowering shoot of a plant species. The species are labelled A, B, C and D. The required equipment and a magnifying glass microscope are on your bench as well.

Analyse the given objects A, B, C and D morphologically. Answer the following questions with the help of the answer code and fill in the corresponding code numbers in the table below.

1 (2.1) Which floral diagram (D 1 - D 6) belongs to which species? Fill in (4 pts) your decision (code number) in the table at the end of this task.



use **D1** **D2** **D3** **D4** **D5** **D6**
as code numbers

1 (2.2) Which family name belongs to which species? Fill in the corresponding code number in the table.

Code numbers:

- | | | |
|------------------------|-------------------------|--------------------|
| 1. <i>Asteraceae</i> | 4. <i>Lamiaceae</i> | 7. <i>Rosaceae</i> |
| 2. <i>Brassicaceae</i> | 5. <i>Oleaceae</i> | |
| 3. <i>Fabaceae</i> | 6. <i>Ranunculaceae</i> | |

1 (2.3) Which species have which kind of fruit (F 1 - F 7)? Fill in the (4 pts) corresponding code number in the table.

Code numbers:

- | | |
|-----------------------|--|
| F 1. Achene | F 4. Fruit composed of four one-seed nutlets |
| F 2. Follicular fruit | F 5. Aggregate of nutlets |
| F 3. Pulse | F 6. Stone fruit, Drupe |
| | F 7. Berry |

1 (2.4) Which species have which inflorescence (B 1 - B 7)? Fill in the (4 pts) corresponding code number in the table.

- | | |
|----------------|----------------------|
| B 1. Spike | B 4. Thyrsus |
| B 2. Capitulum | B 5. Bunch, Raseme |
| B 3. Panicle | B 6. Solitary flower |
| | B 7. Volute |

1 (2.5) Which species have which leaf arrangement? Fill in the corresponding code number in the table.

Code numbers:

- P 1. alternating
P 2. decussate
P 3. distichous

Table

	A	B	C	D
Diagram				
Family				
Fruit				
Inflorescence				
Leaf arrangement				

5.2.2 Laboratory 2: Physiologically - orientated

Task 1: Model experiment referring to the respiratory chain

(23 pts)

Introducing remarks:

Electron transfer through several redox systems to the terminal acceptor as it happens for instance in the endoxidation of the respiratory chain can be simulated in a model experiment in a test tube, following the so-called BAUMANN experiment.

Add Fe^{2+} to a solution of sodium acetate containing the amino acid cysteine, shake it for a short time and write down the colour immediately and after some minutes. Shake the mixture again and write down the colour immediately and after some minutes etc.

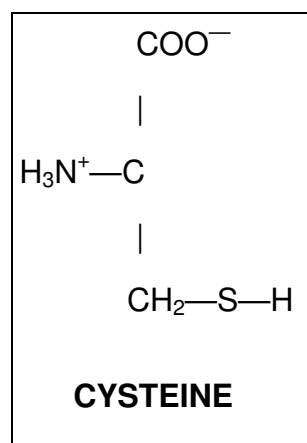
Information: Cysteine forms a coloured compound with Fe^{3+} , whereas Fe^{2+} does not.

I: Preparation of the experiment

The following solution is necessary for the experiment:

10 ml of aqueous 0,1 M sodium acetate solution containing the amino acid cysteine in the concentration of 41 mM (=solution a).

It should be produced by mixing 0,2 M sodium acetate solution (solution b) with 82 mM cysteine solution (solution c).



2 (1.1) How many mg sodium acetate (waterless) have to be weighed for (3 pts) 00 ml of the 0,2 M sodium acetate solution? For calculation use the following

atomic masses: H:1; O: 16; C: 12; Na: 23.

Result (no decimals)

1640 mg

2 (1.2) How many g cysteine have to be weighed for 100 ml of the 82 mM (3 pts) cysteine solution? For calculation use the following atomic masses:

H: 1; O: 16; C: 12; S: 32; N: 14.

Round off the result (no decimals):

1 g

Obtain solution b (6 ml), solution c (6 ml) and the required Fe SO₄ from the assistant and start the experiment.

II: Procedure

2 (1.3) Mix solution b and solution c in the correct volume ratio to get (2 pts) 10 ml of the required solution a.

The use of a pipette aid is obligatory.

Present your experiment and the remaining solutions immediately for judgement (score credit) to the laboratory advisor. Leave your pipette aid with the laboratory advisor.

If you have made a mistake in the procedure you can get solution b and solution c again. In this case you cannot score in (3).

2 (1.4) Add FeSO₄ to the solution a, plug the test tube with a stopper and (1 pt) shake it strongly. Which colour does the solution have after shaking? Mark with a cross.

- ☐ red
- ☐ white
- ☐ green
- ☐ blue-violet

2 (1.5) Leave the test tube open (without stopper) for 1-2 minutes. Which (1 pt) colour does the solution have after this time? Mark with a cross.

- ☐ red
- ☐ white
- ☐ green
- ☐ blue-violet

2 (1.6) How often can the colour change be achieved by shaking?

(1 pt) **Mark with a cross.**

- ☐ once
- ☐ up to 5 times
- ☐ more than 6 times
- ☐ not repeatable

III: Evaluation of the experiment:

2 (1.7) Which component(s) is (are) used up in the experiment?

(1 pt) **Mark with a cross**

- ☐ iron and oxygen
- ☐ cysteine and oxygen
- ☐ sodium acetate
- ☐ oxygen
- ☐ cysteine
- ☐ none

2 (1.8) The redox potential E_0 of the system $\text{Fe}^{3+} / \text{Fe}^{2+}$ is + 0,77 Volt.

(1 pt) **Which redox potential does an e^- donator of this system have?**

Mark with a cross.

- ☐ greater than + 0,77 V
- ☐ less than + 0,77 V
- ☐ equal to + 0,77 V

2 (1.9) Which value does the redox potential E_0 of the e^- terminal

(1 pt) acceptor have in the performed experiment? Mark with a cross.

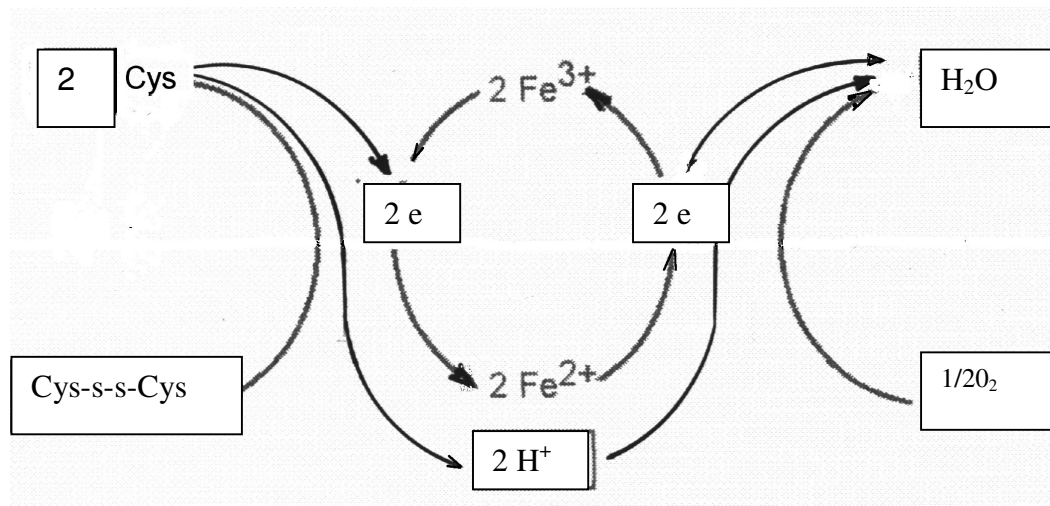
- ☐ greater than + 0,77 V
- ☐ less than + 0,77 V
- ☐ equal to + 0,77 V

2 (1.10) Which enzyme in the respiratory chain is analogous to the Fe-system in the BAUMANN experiment? Mark with a cross.
(1 pt)

- ☐ NADH ubiquinone reductase
- ☐ cytochrome oxidase
- ☐ ATP synthase
- ☐ ubiquinone cytochrome c reductase

2 (1.11) Complete the incomplete schematic drawing of the electron transfer in the BAUMANN experiment.
(8 pt)

You can score 1 or 2 points per correct item in .



Task 2: Imbibition movements of the awns in *Avena sterilis*

(17 pts)

Introducing remarks:

The basis for each imbibition movement is the swelling capability of wall substances. If imbibition occurs in all directions equally, it is called isotropic; if one direction is preferred, the imbibition is called anisotropic. Curvatures can be the result of a substantially different wall structure as well as a different texture – even in homogeneous material. Texture is the arrangement of fibrils in the wall layers. The movements to be observed in the proposed experiment are caused by different textures. Your object is distinguished by two conspicuous awns. You have to examine their movements. A magnifying glass (direct light and transmitted light) is available.

I. Procedure:

2 (2.1) Observe the movement of the awns after moistening.

(2 pts) **Put the air-dry spikelets together with the awns on a piece of paper and sprinkle the awns with water (use the sprayer) and observe (with the help of the magnifying glass as well) your object. Which reactions can you observe? Mark with a cross.**

- ☐ No reaction
- ☐ Both of the awns show a synergic rotation based on decoiling.
- ☐ The awns do not touch each other.
- ☐ Both of the awns show a synergic rotation based on spiralisation. Occasionally they hook together and sometimes start 'saltatory movements'.
- ☐ Both of the awns show a synergic rotation based on decoiling. Occasionally they hook together and sometimes start 'saltatory movements'.
- ☐ The movement of one awn is based on decoiling, the movement of the other one on spiralisation.

2 (2.2) Observe the awn movement when drying.

(2 pts) **Let the awns dry after the experiment (switch both lamps on full power and use them for heating). It can take 10 to 15 minutes until you can see a reaction. Observe your object. Which of the following statements is correct? Mark with a cross.**

- ☐ Both awns show a synergic rotation, which is based on spirali-
sation, occasionally they hook together, and sometimes start
'saltatory movements'.
- ☐ Both awns show a synergic rotation, which is based on
decoiling. The awns do not touch each other.
- ☐ Both awns show a synergic rotation, which is based on
spiralisation.
- ☐ The awns do not hook together.
- ☐ Both awns show a synergic rotation, which is based on
decoiling, occasionally they hook together, and sometimes
start 'saltatory movements'.
- ☐ The movement of one awn is based on decoiling, the
movement of the other awn is based on spiralisation.

II. Evaluation

2 (2.3) Texture and imbibition of cellulose

(3 pts) **On your bench there are two pieces of typing paper. According to the manufacturing process the cellulose fibres are essentially longitudinally orientated.**

2 (2.3a) Give the paper a good spray of water from both sides and decide

(1 pt) **after about 5 minutes whether the paper preferably extents in the longitude of the fibre (A), cross to the fibre direction (B) or in both directions equally (C). Mark with a cross.**

- ☐ A
- ☐ B
- ☐ C

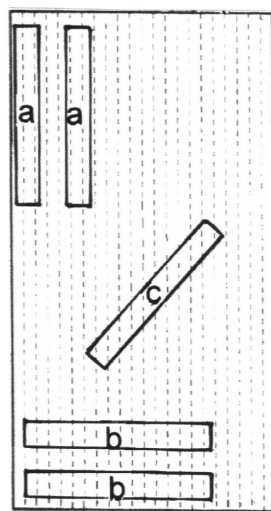
2 (2.3.b) What is the maximum extension given in percent?

(2 pts) **Fill in the result:** 2,3 %

2 (2.4) Perform a mental experiment.

(8 pts) **In figure 1 the fibre direction of a piece of paper is given in dotted longitudinal lines. Stripes of paper are labelled a, b and c. Imagine: The stripes given in figure 1 are cut out, moistened and clued together identically as shown in column 1 of the table 1. For drying the stripes are hung up at the upper end (*).**

Draw the expected form modifications as a side view in column 2 (table 1).



	Column 1	Column 2
A	$\begin{array}{c} a + a \\ \quad \end{array}$	*
B	$\begin{array}{c} b + b \\ \quad \end{array}$	*
C	$\begin{array}{c} b + a \\ \quad \end{array}$	*
D	$\begin{array}{c} a + b \\ \quad \end{array}$	*
E	$\begin{array}{c} a + c \\ \quad \end{array}$	*

1P

1P

1P

1P

1P

2 (2.5) Link the observations regarding the awn movement to the result

(2 pts) **of the (4) pilot experiment (mental experiment).**

Which texture are the movements of the awns based on?

Mark with a cross.

- ☐ (aa)
- ☐ (bb)
- ☐ (ba)
- ☐ (ab)
- ☒ (ac)

5.2.3 Laboratory 3: Genetically - microbiologically oriented

Task 1: Isolation of bacterial DNA

(12 pts)

Introducing remarks:

DNA should be isolated from cells of a 5-day-old liquid culture of the bacterium *Escherichia coli* K 12. Various biochemical steps have to be done in a certain order under different temperatures and incubation times. The tasks are: (1) to determine the correct order of these steps, (2) to assign a specific temperature and incubation time to these steps and (4,5,6) to select the correct substances or solutions for the relevant steps.

3 (1.1) Determine the correct order of the steps. Write down the code letters of the following steps in the correct order.
(4 pts)

- A winding up the DNA
- B disruption of the cytoplasmic membrane
- C precipitation of the uncleaned DNA
- D disruption of the bacterial cell wall

Order: D, B, C, A...

3 (1.2) Assign one of the temperatures (T) and incubation times (I) given below to each of the following steps.
(3 pts)

- Disruption of the cytoplasmic membran T: .+ 60⁰ C I: 2 min..
- Precipitation of the uncleaned DNA: T: . 0⁰ C I: 30 sec
- Disruption of the bacterial cell wall: T: . +37⁰ C. I: 10 min.

Incubation time: 30 sec, 2 min, 10 min, 15 min, 30 min

Temperature: 0⁰ C, room temperature, +37⁰ C, +60⁰ C, +100⁰ C

Realize the following steps in the order you have determined.

3 (1.3) STEP A: Winding up the DNA

(2 pts) **Try to wind up the precipitated DNA strands with a wooden rod.**

Ask the laboratory advisor to score the result.

(The laboratory advisor is going to sign.)

- ☐ DNA could be wind up
- ☐ DNA could not be wind up

3 (1.4) STEP B: Disruption of the cytoplasmic membrane.

(1 pt) **To disrupt the cytoplasmic membrane you have to add 0.2 ml of one of the following solutions.**

Mark the correct answer with a cross.

- ☐ HCl (1 M)
- ☐ acetone (pure)
- ☐ glycerine
- ☐ sodiumdodecylsulphate solution (25 %)
- ☐ NaOH (1 M)

3 (1.5) STEP C: Precipitation of the uncleaned Dann

(1 pt) **For the precipitation of the uncleaned DNA the lysate has to be overlayed with a liquid. Select the correct one from the following liquids and mark it with a cross.**

Realize the overlaying in the following way: Let the fluid from a beaker slowly run down the wall of the test tube. The well-perceptible whitish strands which separate slowly from the aqueous phase after overlaying predominately consist of chromosomal DNA.

- ☐ ethanol (96%, denatured)
- ☐ double distilled water (aqua bidest.)
- ☐ HCl (1 M)
- ☐ NaOH (1 M)
- ☐ glycerine

3 (1.6) STEP D: Disruption of the bacterial cell wall

(1 pt) To disrupt the bacterial cell wall a tip of a spatula of one of the following substances has to be added to a culture of *Escherichia coli* K12 (in a test tube).

Mark the correct substance with a cross.

- ☐ penicillin
- ☐ lysozyme
- ☐ urea
- ☐ CaCl₂
- ☐ trypsin

Task 2: Investigation of the mitosis in the root meristem of a specimen of the *Magnoliopsida* (*Dicotyledoneae*)
(25 pts)

Introducing remarks:

Young roots were harvested from plants grown in sand and differently treated:

Treatment (1): Incubation (for 4 hours) in the aqueous solution of a substance that inhibits chromosome movement

Treatment (2): Incubation (for 24 hours) in ethanol/glacial acetic acid (3:1)

Treatment (3): Incubation (for 24 hours) in a saturated solution of carmine in 45 % acetic acid

Treatment (4): Heating in a boiling water bath for 5 minutes

On your bench you can find two different samples.

Sample I: Root tips only submitted to treatments (2) - (4)

Sample II: Root tips submitted to treatments (1) - (4)

3 (2.1) Compression preparations

(2 pts) Make compression preparations of the samples I and II in which as many cells as possible are in the cell cycle, lying in a

monolayer and flattened in such a way that the chromosome arms are pushed into a focus layer but are not squash-damaged.

Apply a technique you are familiar with or follow the procedure given below.

- Separate the apical meristem of a root in a droplet of 45 % acetic acid on the slide.
- Fragmentate the tip by disteering and slight squashing.
- Cover it with a cover glass.
- Disrupt the tissue clumps by gentle hints with the blunt end of a pair of forceps held upright down on the cover glass.
- Adjust the amount of liquid so that the cells glide but do not float.

In case of insufficient spreading add 45 % acetic acid; slightly lift the cover glass with a needle and avoid shifting of the cover glass by a pair of forceps.

When spreading is sufficient the preparate is covered with a tissue and a gentle pressure by a finger is applied in order to flatten the cells. Attention! The danger of squash-damaging is particularly high when the cells are well spread.

Present your preparates (under the microscope) to the laboratory advisor for quality control and scoring at any time.

3 (2.2) Scetches of different cell stages

(8 pts) **Prepare scetches of the stages of the cell cycle using sample I (a), (b), (c), (e), and (f) and a metaphase-like figure of sample II (d) on the following blank page. Note the list of the stages of the cell cycle below. Draw a chromatide and a daughter chromosome only with a line but otherwise corresponding to your preparate; unclear regions may be circled with dotted lines.**

Stages of the cell cyle:

- (a) interphase (1 pt)
- (b) middle prophase (1 pt)
- (c) metaphase plate in side view (3 pts)
- (d) metaphase-like figure of sample II seen from above (1 pt)
- (e) late anaphase or early telophase (1 pt)
- (f) late telophase (1 pt)

Blank piece of paper (task 2). Sketches of cell stages.

3 (2.3) Detailed drawings of chromosomes

(10 pts) **Draw two metaphase chromosomes from sample I, true to nature, and label the drawing.**

3 (2.3a) Draw two significantly different chromosomes.

(2 pts)

3 (2.3b) Characterize the drawn chromosomes by using capital letters according to the following terms.

(3 pts)

A: metacentric

B: acrocentric

C: dicentric

D: acentric

E: satellite chromosome

3 (2.3c) Label the parts of the larger chromosome using the code numbers of the following terms.

(5 pts)

1. chromosome arm

5. centrosome

2. chromatide arm

6. satellite

3. daughter chromosome

7. nucleolus organisator region

4. centromere

8. nucleosome

Labelled drawings (task 3):

3 (2.4) Chromosome number

(1 pt) **Indicate the chromosome number of sample I.12.....**

3 (2.5) Ploidy level

(1 pt) **Indicate the ploidy level of sample I. Consider that a distinctive chromosome type occurs only once in the haploid chromosome set. Mark with a cross.**

☐ haploid (1 n)

☐ diploid (2 n)

☐ triploid (3 n)

- ☐ tetraploid (4 n)

3 (2.6) Effects of the treatments (1) - (4) (see introducing remarks p.7)

(1 pt) **Match the treatments (1) - (4) to the effects you achieved. Write down the numbers of the treatment in the corresponding circles.**

- ☐ inhibition of the centromere fission
- ☐ inhibition of tubuline association
- ☐ fixation of protoplasmic structures
- ☐ shortening of the chromosomes
- ☐ staining of the chromatine
- ☐ staining of the nucleus spindle
- ☐ destruction of the middle lamella
- ☐ destruction of the primary walls

3 (2.7) Treatment (1)

(1 pt) **Which substance was used in the treatment (1)?**

Mark with a cross.

- ☐ colchicine
- ☐ bromouracile
- ☐ α - amanitin
- ☐ actinomycine
- ☐ mitomycine

3 (2.8) Which aim was followed by J.H. Taylor (1958) when he applied

(1 pt) **the same substance? Mark with a cross.**

- ☐ production of polyploid plants
- ☐ verification of the hypothesis of semiconservative replication in eukaryotes
- ☐ inhibition of mRNA synthesis
- ☐ inhibition of translation
- ☐ investigation of the cytomorphogenesis in *Desmidiacea*

5.2.4 Laboratory 4: Ethologically orientated

Task: Experimental investigation of triad grasp, forceps grasp

(35 pts) **and tool (grasp of man as functions of primate hands).**

Introducing remarks:

During the evolution of primates the grasping hand has developed in a way that it can be used for locomotion as well as for manipulation of objects. Therefore a differentiated co-operation of hand, eye and brain is necessary.

For grasping or splitting tiny food items the ability to manipulate is particularly important (fine tuned motor function) to get sufficient food energy per unit of time and without foreign bodies.

The required equipment and objects are situated on your bench: 2 Petri dishes, a pair of angle forceps, sea sand (sieved), 20 sunflower seeds and a stop-watch. Precision scales are available for the weighing procedure.

Task:

4 (1) Move your thumb and index finger of one hand without any optical control against each other. In which area do they touch each other? Mark with a cross.

(2 pts)

- ☐ inner side of the thumb and outer side of the finger
- ☐ outer side of the thumb and inner side of the finger
- ☐ outer side of the thumb and outer side of the finger
- ☐ inner side of the thumb and inner side of the finger

4 (2) How is this co-ordination possible? Mark with a cross.

(2 pts)

- ☐ by learning
- ☐ by ontogenetic maturing
- ☐ by the construction of the skeleton

- ☐ by neuronal decisions
- ☐ by flexor and extensor muscles

4 (3) Figure 1 shows the print of a man's hand (Dr. Lucius), figure 2 (2 pts) shows the print of a chimpanzee's hand. How are the recognizable structures on the surface of the inner hand of man and chimpanzee called? S K I P P E D.

- ☐ capillary ridges in both
- ☐ tactile fields in both
- ☐ prehensile relief of chimpanzees, papillary ridges of man
- ☐ papillary ridges in both

Print of a chimpanzee's hand (figure 1).



Print of a Dr. Eckhard R. Lucius's hand (figure 1).



4 (4) Investigation of triad grasp, forceps grasp and tool grasp of
(29 pts) **man: Arrange 20 sunflower seeds in the sand-filled Petri dish I in such a way that they lie flat, distributed over the whole dish, and do not touch each other.**

4 (4a) Grasp each seed one after the other with your thumb, index
(5 pts) **finger and middle finger at the same time (triad grasp). All 20 seeds are put individually one after the other in the empty Petri dish II. Measure the required time in minutes and seconds (e.g. 0:49 minutes) with the help of a stop-watch. Smoothen the sand in dish I, transport all the seeds with the help of a pair of forceps from dish II to dish I and weigh the sand (on a filter paper, in mg) that has been transported from dish II when grasping. Repeat the experiment 10 times, write down all data and fill in the results in the table on page 11.**

4 (4b) Grasp each seed from the smoothened sand with your
(5 pts) **thumb and index finger only (forceps grasp). Repeat the procedure from task (4.1).**

Determine the time and the transported amount of sand in 10 experiments and fill in the data in the table on page 11.

4 (4c) Prepare the following experiment as described under (4.1).
(5 pts) **However, this time a pair of forceps should be used for picking up the seeds (tool grasp). Repeat the experiment 10 times, measure the time and amount of sand and fill in the results in the table on page 11.**

4 (4d) Decide in which of the experiments the smallest amount of sand
(4 pts) **(on average) was transported together with the seeds. Mark with a cross.**

- ☐ smallest amount of sand in (4.1) (triad grasp)
- ☐ smallest amount of sand in (4.2) (forceps grasp)

☐ smallest amount of sand in (4.3) (tool grasp)

4 (4e) Calculate the mean and the standard deviation ($\bar{x} \pm s$) of (10 pts) **n = 10 values for the transported amount of sand and the required time in the experiments (4.1) to (4.3). Fill in the results in the table on page 11 and evaluate the significance of the transported sand with the help of the t-test. Mark below which of the differences is (are) significant and which is (are) insignificant.**

experiment 1 versus experiment 2 ☐ significant ☐ insignificant difference

experiment 1 versus experiment 3 ☐ significant ☐ insignificant difference

experiment 2 versus experiment 3 ☐ significant ☐ insignificant difference

Table

	experiment 1		experiment 2		experiment 3	
exp. no.	time (sec)	sand (mg)	time (sec)	sand (mg)	time (sec)	sand (mg)
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
sum						
mean						

standard deviation						
--------------------	--	--	--	--	--	--

Evaluation:

(1.) experiment 1: experiment 2 (sand)

$$t = \text{_____}; \quad p$$

(2.) experiment 1: experiment 3 (sand)

$$t = \text{_____}; \quad p$$

(3.) experiment 2: experiment 3 (sand)

$$t = \text{_____}; \quad p$$

Appendix to the practical test - ethologically orientated

Test for differences in mean between two measuring series.

It has to be tested whether the difference between the results of the three experiments (amount of sand in mg) is statistically significant.

$$n_1 =$$

$$n_2 =$$

$$n_3 =$$

$$\overline{x}_1 \pm s_1 =$$

$$\overline{x}_2 \pm s_2 =$$

$$\overline{x}_3 \pm s_3 =$$

Degrees of freedom

$$m_1 = n_1 + n_2 - 2$$

$$m_2 = n_1 + n_3 - 2$$

$$m_3 = n_2 + n_3 - 2$$

(1.) experiment 1: experiment 2

$$s_D = \sqrt{s_1^2 + s_2^2} =$$

$$t = \frac{|\overline{x}_1 - \overline{x}_2|}{s_D} =$$

(2.) experiment 1 : experiment 3

$$s_D = \sqrt{s_1^2 + s_3^2} =$$

$$t = \frac{|\overline{x}_1 - \overline{x}_3|}{s_D} =$$

(3.) experiment 2 : experiment 3

$$s_D = \sqrt{s_2^2 + s_3^2} =$$

$$t = \frac{|\overline{x}_2 - \overline{x}_3|}{s_D} =$$

t- table (acc. to STUDENT)

	probability p for t
--	---------------------

Individual Results and Medals Awarded

m	0,05	0,01	0,001
18	2,10	2,88	3,92